When is an AI system's response to a benchmark item predictable, when is it not, and what sets the error?

A decomposition, bounds and a method league for the Brier learning curve under the PAIEC protocol, with a confirmatory re-analysis on fresh seeds and an audit of the training release

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Abstract

When can the correctness of an AI system's answer to a benchmark item be predicted, and what sets the error? We study this for the PAIEC protocol (Brier learning curve over 0–31 revealed labels per subject–benchmark pair) on 142 public benchmarks held out benchmark-wise, with a text-free hierarchical predictor as the submitted model and a confirmatory re-analysis on fresh seeds with rules frozen before the runs. (i) An exact three-term learning curve (prior, sampling, split) with all moments taken from the data and nothing fitted matches the beta-binomial baseline within 5% at every budget on fresh seeds (Design B; rule: $\pm 10\%$); our stated expectation of $\leq 3\%$ was missed at budget 31. (ii) Bounds locate the reachable gains: a pair-level floor for text-free predictors, a benchmark-location oracle at budget 0, and a per-benchmark response-noise floor with median 0.049 (IQR 0.011–0.088) on the 62 benchmarks with at least 100 repeated cells. (iii) In the confirmatory runs (benchmark-cold Design A and subject-cold Design B) nine of ten frozen hypotheses passed and one failed (H4, below). With one subject per benchmark the submitted predictor is neither shown better than nor equivalent to the beta-binomial baseline (-0.0004 ± 0.0004 ALC in both designs, grey zone); with three subjects sharing the labels it is better by 0.0049 ± 0.0007 (both designs). Subject identity shows a small effect (2.1–2.6 SE); the equivalence hypothesis failed. In a method league over all 10 folds, with one subject per benchmark none of the smoothing, IRT, matrix-factorisation, kNN or stacking families is distinguishable from the submitted model; gradient boosting and the unsmoothed baselines are worse, and from two subjects on the submitted model is best or tied-best in every regime. (iv) An audit of the training release (9 findings, 17 evidence packs, 3 patches) accompanies the submission. Code and report: https://github.com/christian140903-sudo/christian140903-sudo/tree/claude/technosignatures-archive-analysis-nxoino/paiec-2026.

1 Introduction

When is an AI system's response to a benchmark item predictable, when is it not, and what sets the error? Our thesis is that, under the PAIEC protocol and for predictors that use no item text, the error is *information-limited, not model-limited*: it is set by what the protocol reveals (labels, subjects of the same benchmark sharing them, the item split, the item count), not by the choice among estimators that use the revealed labels. We test this; we do not assume it. Item response models evaluate language models on fewer items [Maia Polo et al., 2024, Kipnis et al., 2025, Vivek et al.,

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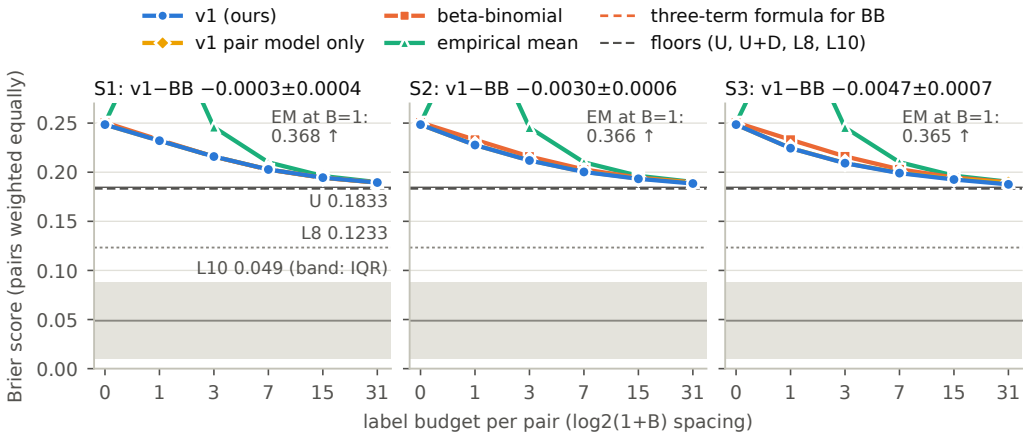


Figure 1: The map (discovery set): Brier per budget for EM, BB, the pair-level model and the submitted model at one, two and three subjects per benchmark, with the pair-level floor U and the three-term curve for BB (dashed, nothing fitted). Panel titles: submitted − BB ± SE ("v1" in the panels = the submitted predictor along the v1 path, Appendix B). Grey: floor U (thin: $U + D$), oracle L8, repeat-noise floor L10 (median, band = IQR). Data: Table 10.

2024], capability measures predict performance across models [Ruan et al., 2024, Owen, 2024], and amortised, text-based item parameters are the organisers' own line [Truong et al., 2025]; missing, to our knowledge, is what a label-budget protocol lets *any* text-free predictor achieve, how fast the remaining error falls with labels, and where it fails. PAIEC [Stanford AIMS, 2026b] scores the Brier score [Brier, 1950] over subject–benchmark pairs as an area under the learning curve (ALC) over budgets 0, 1, 3, 7, 15, 31 [Guyon et al., 2011], on benchmarks disjoint from training.

Contributions. (a) An exact per-pair decomposition and a ladder of oracle bounds down to the per-benchmark response-noise floor (§3). (b) A three-term learning curve for posterior-mean predictors derived from the protocol, tested with no fitted parameter on fresh seeds, and its measured limit, pooling. (c) Protocol dependence as tested statements (Table 4). (d) A method league under one protocol and a hypothesis table reported as it fell, including a failed hypothesis. (e) A data-quality audit of the training release with evidence packs, patches, a curation metric and a checking tool (§7).

2 Task, protocol and data

At each budget the platform calls `predict(input, labeled)` per distinct `[subject, item]`; the subject is eight visible string fields, the benchmark an anonymous stable `benchmark_id`. Per pair with ≥ 80 items the items are split 50/50 into acquisition and evaluation pools; labels are revealed at random, one recorded response per item, nested over $B \in \{0, 1, 3, 7, 15, 31\}$; the pair score is the Brier score over all recorded responses of its evaluation items, and $\mathrm{ALC} = 0.1\, B_0 + 0.2\,(B_1 + B_3 + B_7 + B_{15}) + 0.1\, B_{31}$. The streaming driver [Stanford AIMS, 2026c] passes one `labeled` set per checkpoint to all targets of a run. A subject *is* its field dictionary: pairs with identical fields are indistinguishable, and every predictor, the organisers' baselines included, counts their labels together ($\approx 2\%$ of pair–budget rows get more "own" labels than B; mean surplus at $B = 31$: $+0.6$, $+1.2$, $+8.9$ with two, three, thirty subjects per benchmark, none with one). How many subjects of one benchmark share a run on the platform is not public.

The only training source is the Measurement Data Bank [Stanford AIMS, 2026a], revision `54074453`: 156 public binary benchmarks with grades, 31,389,917 subject–item rows, 68,268,267 responses; no other data, no text, no language model. Responses are used as released (7 `lower_is_better` benchmarks unflipped); two groups are excluded on the basis of the audit (§7): 64 bird_sql configurations graded 0 on all 360 questions and 12 swebench_live subjects with only successes. Pairs with ≥ 80 items occur in 142 benchmarks (12,918 pairs, median benchmark 13, two benchmarks hold 59%); 52% of the pairs have no `normalized_name`.

3 Theory: what sets the error

Exact decomposition. If a pair has success rate R on its evaluation responses and a predictor issues probabilities q_i with response-weighted mean $\bar{q}$ (Appendix A, A.1),

$$\text{Brier}_B = (\bar{q} - R)^2 + R(1 - R) + G_B, \tag{1}$$

with item gain $G_B \leq 0$ non-zero only with item-level information. $R(1-R)$ is the floor of every pair-constant predictor; its ALC-weighted mean on the test pairs is $U = 0.1833$. Its Murphy decomposition [Murphy, 1973, Gneiting and Raftery, 2007] shows for the submitted model (binned, $S = 3$) reliability ≤ 0.0006 at every budget and resolution 0.0022 of $\text{Var}(R) = 0.0667$ at budget 0: budget 0 is a resolution problem, not a calibration problem.

Learning curve of a posterior-mean predictor. For $q_B = (k + \kappa\mu)/(B + \kappa)$ with fixed prior strength κ ($\alpha + \beta$ of a Beta prior), N items per pair split into halves A and E, Appendix A.2 derives in eight steps

$$\text{E}(q_B - R_E)^2 = \frac{B\,(U - (B-1)F) + \kappa^2 V_0}{(B+\kappa)^2} + \frac{B}{B+\kappa}\,D, \qquad \text{E}(q_0 - R_E)^2 = V_0, \tag{2}$$

with prior error $V_0 = \text{E}(\mu - R_E)^2$, split term $D = \text{E}[4\,\text{Var}_i(p_i)/(N-1)]$ (the acquisition half is not the evaluation half) and finite-population term $F = \text{E}[\text{Var}_i(p_i)/(N_A - 1)]$. It is the standard Bayesian mean-squared error of a posterior-mean predictor [Morris, 1983, Efron and Morris, 1975], applied to the competition protocol with a split term; U, V_0, κ, D, F are moments of the data, not fitted (discovery: 0.1833, 0.0673, 2.73, 0.00138, 0.00070; N median 1,000, minimum 80). Disclosure: the split term was added after a 13% shortfall at budget 31 on the discovery set, before any confirmatory run (A.2). Consequences: $\kappa^* = U/V_0$ minimises every budget's error of the two-term form (2.76 with the split term; for a well-specified Beta prior $U/V_0 = \kappa$ identically, so "$\kappa^* \approx \kappa_{\text{ML}}$" is a specification test); the $1/(B+\kappa)^2$ shape puts the loss early (budget 31 carries about 2%); even a perfect prior leaves a pair-level floor for text-free predictors of U + sampling + split $= 0.1948$ ALC, and the sampling term alone (0.0107) exceeds the whole model–baseline gap at three subjects (0.0047).

Validity limit: pooling. The submitted model is *not* a fixed-κ predictor: its benchmark location is updated from the other subjects' labels, so its effective V_0 depends on S and B. With its own budget-0 excess, Eq. 2 overstates its error at one label by $+17.6\%$ (three subjects) and $+78.4\%$ (thirty) on the discovery set, vanishing by budget 31. We therefore describe pooling as a budget- and subject-count-dependent reduction of the prior error, $V_0(S, B)$, and test it as H2b; the formula is not a general statement about learning curves. The identity collision makes $S = 1$ the clean test regime.

Bound ladder and noise floor (Fig. 4 in Appendix D; A.5). At budget 0 the constant scores 0.2500, the submitted model 0.2485, and a leave-one-out oracle of the benchmark location 0.2067: benchmark location is the largest lever at budget 0 (65% of $\text{Var}(R)$), out of reach for text-free predictors under our rules (ability alone adds 3–6%). The pair-rate oracle gives $U = 0.1833$: the within-pair term is 73% of the constant predictor's Brier score of 0.25 (92% of $\text{Var}(y) = 0.199$); rate plus leave-one-out item difficulty removes a third of it ($0.1833 \to 0.1233$), and the cell-rate oracle k/n leaves 0.0065 (L9, biased low: cells with one response contribute 0). The bottom is response noise: for cells with $n \geq 2$ responses, $k(n-k)/(n(n-1))$ is unbiased for $p(1-p)$ (L10). On the 62 benchmarks with repeated responses the per-benchmark noise floor has median 0.049 (IQR 0.011–0.088); the answer-weighted mean 0.044 is dominated by one benchmark, and L10 is a subset bound because repeats exist only where curators recorded trials [cf. Madaan et al., 2024, Miller, 2024, Northcutt et al., 2021, Gema et al., 2025].

4 Method

The submitted predictor models $\text{logit}\, r_{sb} = c_b + a_b m_s + v_{sb}$ with $c_b \sim \mathcal{N}(\mu_c, \tau_c^2)$, $a_b \sim \mathcal{N}(\lambda, \tau_\lambda^2)$, $v_{sb} \sim \mathcal{N}(0, s_s^2)$: m_s is the subject's ability from training (a ridge regression on visible metadata for unseen names); location c_b and slope a_b are always cold, learned only from the labels of the run. Hyperparameters come from Gaussian marginal likelihood on inner benchmark folds [MacKay, 1992]. At test time the labels are filtered by `benchmark_id` and grouped by the eight fields; the posterior over $(c_b, a_b, v_1 \ldots v_G)$ is a Laplace approximation solved in $O(G)$, the probability its probit-approximated mean. An item stage adds a Laplace posterior for an item offset $d_i \sim \mathcal{N}(0, \tau_i^2)$ from

other subjects' labels on the same item ($\tau_i^2 = 2.55$ from a one-parameter IRT fit [Lord, 2012, Lalor et al., 2016]); without such labels it returns the pair-level value exactly. Platt scaling [Platt, 2000] per label stage is kept at stages 0 and 1 ($0.2527 \to 0.2485$ at budget 0; stage choice in Section 8). One file, standard library only, no network; `predict` never raises: on an exception it falls back to a beta-binomial estimate from the pair's own labels, then to 0.5 (Appendix C).

Why no text: a decision with a bound. At budget 0 the submitted model scores 0.2485 and text-free ability oracles 0.2480–0.2460 (A.5); the benchmark-level oracle shows that benchmark-level text could lower this to at most 0.2067, and the item-difficulty oracle puts the ceiling of item information at -0.054 ALC. That room is what amortised, text-based approaches target [Truong et al., 2025, Zhou et al., 2025, Pacchiardi et al., 2024]; ours is complementary: it states what no text-free predictor can do and how fast the remaining error falls with labels. A text-based item prior was prepared but not completed before the freeze and is not part of the submission (Appendix H).

5 Experimental design

Replica. The offline evaluation re-implements the published protocol from the competition text and `streaming_ingestion.py` (read, not executed): 50/50 split per pair and seed, revealed labels as the first B items of a random permutation of the acquisition half (equal in distribution to the platform's selection sampling), one recorded response per revealed item, exact Brier over the evaluation items (assumptions A1–A5 and the leak regression: Appendix C). We reproduce the organisers' Empirical Mean (EM) and the beta-binomial baseline (BB) exactly in our harness (max $|\Delta q| = 0$).

Designs, regimes, seeds. Design A (benchmark-cold): 10 folds over 142 benchmarks, everything fitted on training benchmarks only. Design B (benchmark- and subject-cold): names additionally split into 5 folds and removed from training. At most 30 test pairs per benchmark (2,142 pairs). Regimes $S = 1, 2, 3$ with equal standing, $S = 30$ as a limit, compact10 (10 subjects share 100 items) as a hypothetical bound for item overlap. The *discovery set* (fold seed 1234, seeds 0–2) is where every hypothesis in Table 2 originated. The *confirmatory re-analysis* (fold seed 200, seeds 200–202) ran with rules frozen before the runs: the plan was fixed at 2026-10-20 20:26:04 UTC (sha256 `e5fd821f...`; full hash and text in Appendix B), before the first confirmatory run started, for the hardened variant of the submitted predictor (deviation noted in Appendix B). It is confirmatory with respect to sampling randomness, not with respect to the benchmark population: all hypotheses originated from the same 142 benchmarks.

Decision rule. All comparisons are paired over the same pairs and seeds, SE cluster-robust over benchmarks [Cameron and Miller, 2015]. *Better*: $\hat{d} + 2\,\mathrm{SE} < 0$; *worse*: $\hat{d} - 2\,\mathrm{SE} > 0$; *equivalent*: $|\hat{d}| + 2\,\mathrm{SE} < 0.001$ ALC (margin fixed in advance); otherwise *grey zone*. Under a null effect a false *better* or a false *worse* verdict each occur with probability 2.3% (4.6% for either); equivalence is undecidable at $\mathrm{SE} \geq 0.0005$ and has 38% power at $\mathrm{SE} = 0.0004$ (45–46% at the SE of 0.00038 observed for H1 at $S = 1$), so a grey zone there is "not decidable". About 20 directional verdicts are produced, so about 0.9 false ones are expected under a global null (the pre-registration stated ≈ 0.5, counting one direction only); no correction applied (Appendix B); league competitors are tuned only on the training benchmarks of their fold.

6 Results

Main result (Table 1, Fig. 1). The confirmatory runs give, submitted predictor against BB, -0.0004 ± 0.0004 ALC with one subject per benchmark in Design A and in Design B (grey zone: neither shown relevant nor shown equivalent; power for equivalence 45–46% at the observed SE of 0.00038), -0.0031 ± 0.0006 with two (Design A) and -0.0049 ± 0.0007 with three (both designs; better), -0.0132 ± 0.0012 with thirty; against EM -0.0347 ± 0.0024 at one subject (both designs). On the discovery set: -0.0003 ± 0.0004, -0.0030 ± 0.0006, -0.0047 ± 0.0007 and -0.0130 ± 0.0013 at one, two, three and thirty subjects. Almost all of the distance to EM is smoothing (BB alone gains 0.034); the model's own gain is a pooling gain, shown only when at least two subjects of a benchmark share the labels.

Hypotheses as they fell (Table 2). Of ten frozen hypotheses nine passed and one failed. *H4 failed:* subject identity shows a small effect (-0.0006 ± 0.0003 at two and -0.0007 ± 0.0003 at three subjects, 2.1 and 2.6 SE); the equivalence hypothesis failed, and "subject knowledge carries little" is

Table 1: Main result per regime. Columns 2–7: **discovery** (Design A, seeds 0–2, 10 benchmark folds, 2,142 test pairs, 142 benchmarks; paired differences ± cluster-robust SE over benchmarks; letters = verdict by the rule of Section 5: b better, w worse, e equivalent, g grey zone). Last two columns: **confirmatory** runs (seeds 200–202, fold seed 200; H1 of Table 2), Design A (benchmark-cold, all regimes) and Design B (benchmark- and subject-cold, S1 and S3). "ours" = the submitted predictor; discovery numbers were computed along the v1 path, which differs from the submitted v2.0-core only in the Platt stages 3 and 7 (Appendix B). Sources: `modell/ergebnisse/cv_A_p2_S1-S2-S3-S30-kompakt10/auswertung.json`, `liga/ergebnisse/liga_vergleich.json` (ALC columns), `konfirmation/hypothesen_tafel.json`.

	Brier ALC (lower is better)				ours − baseline (discovery)		ours − BB (confirmatory)	
regime	EM	BB	pair model	ours	vs BB	vs EM	Design A	Design B
S1 (1 subject/benchmark)	0.2479	0.2131	0.2129	0.2129	-0.0003 ± 0.0004 g	-0.0351 ± 0.0024	-0.0004 ± 0.0004 g	-0.0004 ± 0.0004 g
S2	0.2475	0.2133	0.2106	0.2103	-0.0030 ± 0.0006 b	-0.0372 ± 0.0021	-0.0031 ± 0.0006 b	—
S3	0.2475	0.2134	0.2091	0.2087	-0.0047 ± 0.0007 b	-0.0388 ± 0.0020	-0.0049 ± 0.0007 b	-0.0049 ± 0.0007 b
S30 (limit)	0.2470	0.2135	0.2042	0.2005	-0.0130 ± 0.0013 b	-0.0465 ± 0.0021	-0.0132 ± 0.0012 b	—
compact10 (hypothetical)	0.2504	0.2158	0.2083	0.1968	-0.0190 ± 0.0014 b	-0.0536 ± 0.0028	—	—

kept only with the number (under one fifth of the pooling gain). *H1 passed* (Table 1). *H2a passed* in both designs: the curve matches the baseline's budget dependence within $+1.6$, $+2.0$, $+2.5$, -2.4, -4.2% (Design A) and $+1.6$, $+1.9$, $+2.4$, -2.4, -4.2% (Design B) at budgets 1–31 (rule $\pm 10\%$ met; expectation $\leq 3\%$ missed at budget 31). *H2b passed:* all six pooling steps better; the fixed-κ form overstates the pooled model's error at one label by $+21\%$ (three subjects) and $+88\%$ (thirty), inside the frozen ranges. *H2c passed:* $\bar{\kappa} = 2.735$ against $U/V_0 = 2.725$ (bootstrap SE 0.31). *H3 passed:* 77–78% of the excess over the pair floor lies at budgets 0–3 (lower confidence bounds 0.75–0.76). *H5 passed:* none of the nine mixture, stacking and non-parametric-prior comparisons moved ALC by more than 0.00012 (all equivalent). *H6 passed:* visible non-text item aggregates do not predict benchmark location at budget 0 (leave-one-benchmark-out R^2 -0.048 for R_b, -0.056 for its logit). *H7 passed:* L10 median 0.0488 over 62 benchmarks, reproduced by a second computation with different code. *H8 passed:* the item-stage gain is linear in the number of foreign-labelled items (-0.00009 per item, R^2 0.98) and saturates beyond about 50.

Learning curve: match and limit (Fig. 3, App. D). With all moments estimated from the same test pairs, the predicted budget dependence matches the beta-binomial baseline within 0.8% at every budget on the discovery set and within 5% on fresh seeds (Design B); the two-term textbook form fails by -11% at budget 31. For the pooled model the fixed-κ form overstates the early error (§3). On fresh seeds 77–78% of the submitted model's excess over the pair-level floor lies at budgets 0–3 in S1–S3 (H3; discovery 77–79%; 89% at thirty subjects): the prior and the first three labels decide the score.

Method league and ablations (Table 3, Table 5 in Appendix D; Fig. 2). Over all 10 folds, at one subject per benchmark none of the smoothing, IRT, matrix-factorisation, kNN or stacking families is distinguishable from the submitted model by the decision rule (paired differences $+0.0003$ to $+0.0009$ ALC, none significant, none equivalent); gradient boosting ($+0.0025 \pm 0.0008$) and the unsmoothed baselines (EM, constant) are worse. From two subjects on the ordering is set by pooling: the submitted model has the best mean ALC in every regime, no family is better in any regime (0 of 35 family–regime comparisons), and 23 of the 35 are worse by the rule: gradient boosting throughout, kNN, IRT 1PL and the beta-binomial from two subjects on, IRT 2PL [Lord, 2012, Fox, 2010] from three ($+0.0007 \pm 0.0003$; the nearest family from one to three subjects), stacking [Wolpert, 1992] from thirty, matrix factorisation [Koren et al., 2009] only in compact10 (grey zone up to thirty subjects, $+0.0006$ to $+0.0010$). The organisers' LLM-based BLE baseline is not rebuilt. Among the ablations, the item stage contributes -0.0004 ± 0.0001 at three subjects and -0.0037 ± 0.0006 at thirty; Platt scaling and the benchmark slope are not shown to matter; the stratum stage and the text prior were not evaluated (Appendix H).

Error anatomy and two cases (Figs. 5, 6, Appendix D). *Case 2 (failure): Mind2Web* has a single, unnamed subject (success rate 0.90). The submitted model loses 0.0060 ALC to the beta-binomial smoother, 62% of it at budget 0, where its prior for a subject without metadata sits at 0.47 and the smoother's at 0.52; with no second subject in `labeled` there is nothing to pool, and from budget 7 on both predictors agree within 0.001. Across all 13 single-subject benchmarks the picture is

Table 2: Hypothesis table as it fell (pre-registration `prereg/PRAEREGISTRIERUNG-v2.md`, sha256 e5fd821f…, frozen 2026-10-20 20:26:04Z, before the confirmatory runs). Discovery = seeds 0–2 (Design A); confirmatory = seeds 200–202, fold seed 200, Design A unless marked B. Summary of verdicts: 9 passed, 1 failed. Verdicts are produced by code, self-attested, no second judge; "ours" = the submitted predictor (confirmatory runs: nested Platt-stage choice per fold, Appendix B). *rule met, stated expectation of $\leq 3\,\%$ missed at B31. Source: `konfirmation/hypothesen_tafel.json`.

id, estimand	rule	discovery value	confirmatory value	verdict
H1 ALC(ours) − ALC(BB), S1/S2/S3	S2, S3 better; S1 equivalent or grey zone with power	S1 -0.0003 ± 0.0004 (grey zone), S2 -0.0030 ± 0.0006 (better), S3 -0.0047 ± 0.0007 (better), S30 -0.0130 ± 0.0013	A: S1 -0.0004 ± 0.0004 grey; S2 -0.0031 ± 0.0006 better; S3 -0.0049 ± 0.0007 better (S30 -0.0132 ± 0.0012); B: S1 -0.0004 ± 0.0004 grey; S3 -0.0049 ± 0.0007 better	passed
H2a relative error of the exact three-term formula vs the BB curve, $B = 1$–31, S1, Designs A and B	all five budgets within $\pm 10\,\%$ in both designs (expectation $\leq 3\,\%$)	Design A S1 exact: −0.1 / +0.1 / +0.8 / +0.4 / +0.3 %; Design B not compared	A: $+1.6/+2.0/+2.5/-2.4/-4.2\,\%$, max 4.2 % (within 10 %: yes; within 3 %: no); B: $+1.6/+1.9/+2.4/-2.4/-4.2\,\%$, max 4.2 % (within 10 %: yes; within 3 %: no)	passed*
H2b (i) pooling steps S1→S30 at B1, B3; (ii) fixed-κ formula vs pooled model at S3/B1, S30/B1	(i) 6/6 steps better; (ii) S3/B1 in [+10, +30] %, S30/B1 in [+50, +110] %	(i) 6/6 steps better (B1 0.0488→ 0.0272, B3 0.0327→ 0.0192 from S1 to S30); (ii) S3/B1 +17.6 %, S30/B1 +78.4 %	(i) 6/6 steps better; (ii) S3/B1 $+21.4\,\%$, S30/B1 $+87.7\,\%$	passed
H2c $\bar{\kappa}$ (BB prior strength per fold) vs U/V_0 (S1, B0)	$\lvert\bar{\kappa} - U/V_0\rvert \leq 2\,\mathrm{SE}_{\mathrm{boot}}$	$\bar{\kappa}_{\mathrm{ML}} = 2.731$, $U/V_0 = 2.722$, difference $+0.009$, $\mathrm{SE}_{\mathrm{boot}}$ 0.309	$\bar{\kappa}$ 2.735, U/V_0 2.725, difference +0.010, $\mathrm{SE}_{\mathrm{boot}}$ 0.309 ($\bar{\kappa}$ fitted by moments, as in discovery; Appendix B, deviation 3)	passed
H3 share of our excess at B0–B3 (cluster bootstrap)	lower 95 % bound > 0.70 in S1, S2, S3	S1 0.772 [0.753, 0.788], S2 0.781 [0.761, 0.799], S3 0.785 [0.765, 0.805]; S30 0.892 [0.841, 0.959]	S1 0.767 [0.748, 0.785]; S2 0.778 [0.759, 0.795]; S3 0.783 [0.764, 0.801]; S30 0.894 (reported)	passed
H4 ALC(ours) − ALC(ours without subject identity), uncalibrated	equivalent ($\lvert d\rvert + 2\,\mathrm{SE} < 0.001$) in S1, S2, S3	S1 -0.00048 ± 0.00035, S2 -0.00050 ± 0.00031, S3 -0.00051 ± 0.00028 — all grey zone	S1 -0.0005 ± 0.0003 grey; S2 -0.0006 ± 0.0003 better; S3 -0.0007 ± 0.0003 better (none equivalent)	**FAILED**
H5 mixing / stacking ours with BB; NPEB vs BB	all nine equivalent, none better	nine comparisons, all equivalent, none better ($\lvert d\rvert \leq 0.00030$)	9 of 9 equivalent, 0 better; $\lvert d\rvert \leq 0.00012$, SE ≤ 0.00023 (largest: (b) stacking S2/S3 $+0.00012$; nine values in `hypothesen_tafel.json`)	passed
H6 LOBO-R^2 of benchmark metadata on R_b and logit R_b (ridge)	LOBO-$R^2 \leq 0.05$ for both targets	feature set A: LOBO-R^2 −0.024 (R_b), −0.025 (logit)	LOBO-R^2 −0.048 (R_b), −0.056 (logit); feature set B	passed
H7 L10 repeat-noise floor: median over benchmarks with ≥ 100 repeated cells	median ≥ 0.02; two computation paths within 1 %	median 0.0488 (IQR 0.011–0.088), 62 benchmarks, one computation path	median 0.0488, IQR 0.011–0.088, 62 benchmarks; second path median 0.0488	passed
H8 item-stage gain at B31 vs number of evaluation items with foreign labels	(i) S3 slope < 0, $R^2 \geq 0.8$; (ii) S30 saturation	(i) S3 slope −0.00009/item, R^2 0.995; (ii) S30: 21–50 −0.0157, 51–100 −0.0125, >100 −0.0138 (saturation)	(i) S3 WLS slope −0.00009/item, R^2 0.98; pair slope −0.00008 [−0.00011, −0.00005]; (ii) S30: 21–50 −0.0171, 51–100 −0.0123, >100 −0.0140 (saturation)	passed

the same ($+0.0032 \pm 0.0006$; 12 of 13 positive). *Case 1 (success): NYU CTF Bench* (15 unnamed agent subjects, mean success rate 0.13). With one subject the model already beats the smoother by 0.0038 ALC, mostly at budget 0 (0.246 against 0.268: its prior for an unnamed agent sits closer to the low success rate); with three subjects sharing `labeled` the gain grows to 0.0204 ± 0.0029, and the additional 0.017 sits at budgets 1–15, where each own label comes with two foreign labels of the benchmark. The same predictor gains 0.020 or loses 0.006 depending on how many subjects share the labels, not on the estimator.

Protocol sensitivity (Table 4, Appendix D) and efficiency. What moves the error besides the model: subjects per benchmark sharing labels ($-0.0003 \to -0.0130$ against BB from one to thirty), the split unit (a per-item split makes $G_B \equiv 0$), the item count ($D, F \propto 1/N$: split floor 0.0090 at $N = 80$, 0.0007 at $N = 1{,}000$), the identity collision (formula error for BB up to -8.3% at budget 31 with thirty subjects), and the acquisition policy, which in a one-fold pilot made no difference at three subjects and helped in compact10 (heavy item overlap; full run not evaluated). Two acquisition lemmas hold without measurement (A.7; Rubin, 1976, Kossen et al., 2021). The predictor costs 0.4 ms per prediction with 100,000 labels passed as a fresh copy, standard library only, no API calls; the organisers' BLE baseline is an LLM agent planned at 240 s per question with an LLM API.

Table 3: Method league, all 10/10 folds, discovery seeds, test pairs capped at 30 per benchmark, hyperparameters chosen on the training benchmarks of each fold (inner 75/25 split). ALC at one (S1) and three (S3) subjects per benchmark; Δ = paired difference method $-$ ours (negative = method better; "ours" along the v1 path, Appendix B) $\pm$ cluster-robust SE; letter = verdict by the pre-stated rule (b better, w worse, e equivalent, g grey zone; Section 5). The ablations are in Table 5 (Appendix D). Sources: `liga/ergebnisse/liga_vergleich.json` (ALC, full values rounded once), `liga/ergebnisse/rangfolge.csv` (mean rank), `liga/ergebnisse/vergleich_gepaart.csv` (Δ).

method	ALC S1	ALC S3	mean rank	Δ vs ours, S1	Δ vs ours, S3
ours (v1 path)	0.2129	0.2087	1.0	reference	reference
IRT 2PL	0.2131	0.2094	3.2	$+0.0003 \pm 0.0004$ g	$+0.0007 \pm 0.0003$ w
ours without item stage	0.2129	0.2091	4.0	0.00000 ± 0.00000 e	$+0.00045 \pm 0.00008$ w
stacking (2PL, MF, kNN, BB)	0.2134	0.2094	4.4	$+0.0006 \pm 0.0007$ g	$+0.0007 \pm 0.0005$ g
matrix factorisation (rank 1–8)	0.2138	0.2095	4.8	$+0.0009 \pm 0.0006$ g	$+0.0009 \pm 0.0005$ g
IRT 1PL (Rasch)	0.2132	0.2102	4.8	$+0.0004 \pm 0.0004$ g	$+0.0015 \pm 0.0004$ w
kNN over subjects	0.2135	0.2106	7.2	$+0.0006 \pm 0.0005$ g	$+0.0020 \pm 0.0004$ w
gradient boosting (20 features)	0.2154	0.2113	7.6	$+0.0025 \pm 0.0008$ w	$+0.0027 \pm 0.0005$ w
beta-binomial	0.2131	0.2134	7.8	$+0.0003 \pm 0.0004$ g	$+0.0047 \pm 0.0007$ w
empirical mean	0.2479	0.2475	10.2	$+0.0351 \pm 0.0024$ w	$+0.0388 \pm 0.0020$ w
constant 0.5	0.2500	0.2500	10.8	$+0.0371 \pm 0.0042$ w	$+0.0413 \pm 0.0047$ w

Publishing the subjects per benchmark in the summative run, the split unit and the pool sizes would help.

7 Training-data audit as a contribution to curation

The Measurement Data Bank documents its sources well; we thank the curators. The audit covers the public release only (revision `54074453`; `paiec_baseline` `cc88a959`), is read-only and infers nothing about the hidden test set; where a rule could matter there, it is phrased as a question. Inclusion standard (two implementations with different code, an external witness where one exists, harmless explanations checked, two refutation attempts): Appendix E. Nine findings (Table 8): at revision `54074453` the public baseline loader sees 60 of 156 eligible graded benchmarks, because the Hub lists the first 100,000 of 385,266 files; a patch with a regression test is supplied, and the stated effect is on the language-model baselines' corpus, not on scoring. In bird_sql, 64 of 120 configurations are graded 0 on all 360 questions while the publisher's aggregate lists them at 5.6–40.6%; 12 swebench_live subjects carry only successes (question); reasoning configurations are stored as trials; 117 Winogrande twin pairs collapsed into identical text with opposite answers; 119 models in 13 benchmarks receive identical competition input; plus documentation items. The loader finding was reported to the organisers on 20 October 2026; the 17 evidence packs and 3 patches were prepared in response to their request. Reusable tools released with the code: `mdb_check.py` (schema, keys, duplicates, polarity, loader coverage, repeat statistics; fail-closed; tested on 7 deliberately corrupted copies and an unmodified control), `noise_floor_by_benchmark.csv` (L10 per benchmark) and an identity repair table (67–74% of the 13,371 unnamed subjects carry a usable identifier).

8 Limitations

Simulator versus platform: subjects per run, split unit, pool sizes and the summative subset are not public, so all results are a range over regimes, not a platform score. **Confirmation:** all ten hypotheses evaluated on fresh seeds (nine passed, H4 failed); sampling randomness only, the same 142 benchmarks, and each fold's nested Platt-stage set rather than the package's exact set (Appendix B). **Formula:** fixed prior strength, in-sample moments, $\leq 3\%$ missed at budget 31. **Noise floor:** a subset bound (62 benchmarks). **No text:** a bound, not a measurement of what text achieves. **Polarity:** 7 benchmarks unflipped. **League:** finite grids (IRT prior strength at the upper edge in all folds; extending the grids on two folds improved the inner ALC by at most 0.0004; gradient-boosting edges not checked). **Researcher degrees of freedom:** the item stage was added after seeing discovery results. Platt scaling per label stage: v1 kept the stages whose cross-fitted Brier was lower (B0 -0.0042, B1 -0.0009, B3/B7 -0.00002; point estimates, stage choice on all folds); v2.0-core keeps only stages 0 and 1, the intersection of the nested per-fold selections (stage 3 was selected in 8 of 10 folds, stage 7 in 7 of 10, stages 15 and 31 in at most 2).

9 Conclusion

Under the PAIEC protocol the error of a text-free predictor is information-limited, not model-limited: a three-term curve with nothing fitted matches the baseline's learning curve within 5% on fresh seeds, bounds locate the reachable gains, and among estimators that use the revealed labels, protocol quantities move the attainable score (0.012 ALC for the submitted model from one to thirty subjects) more than the choice among the other six families (0.0050 ALC for the weakest of them, kNN, at thirty subjects); the beta-binomial, which does not pool labels across subjects, trails by 0.0130 there. The audit outputs are offered to the curators as numbers and code.

Use of AI assistants

Code, offline evaluation, data audit and the drafting of this report were developed with AI coding assistance (Claude Code, Anthropic), as permitted by the competition rules; the author directed the work and is responsible for all content. The re-computations and checks described in the appendices were also carried out by AI agents under the author's direction; they are internal checks, not a review by third parties. The submitted predictor uses no language model at runtime.

Reproducibility and licence

Code, fitted state, both packages, harness, audit tools, results and this report's source (`report/src/`): repository folder linked in the abstract (snapshot of version `v2.0.1-paiec`); Appendix F. `make all` rebuilds the cross-validation tables from the data (not timed as one run; logged phases suggest 20–30 minutes on four cores, 16 GiB); `make smoke` runs tests, one fold against a stored reference and a hash check (about 70 seconds, measured on fresh clones). Submitted package: `paiec_v2core_final.zip` (sha256 `3274b095`…, full hash in Appendix F). Data: measurement-db revision `54074453` (gated; none redistributed). Licences: code MIT, report CC BY 4.0.

References

Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950. doi: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2. URL https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.

A. Colin Cameron and Douglas L. Miller. A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372, 2015. doi: 10.3368/jhr.50.2.317. URL https://doi.org/10.3368/jhr.50.2.317.

Bradley Efron and Carl Morris. Data analysis using Stein's estimator and its generalizations. *Journal of the American Statistical Association*, 70(350):311–319, 1975. doi: 10.1080/01621459.1975.10479864. URL https://doi.org/10.1080/01621459.1975.10479864.

Jean-Paul Fox. *Bayesian Item Response Modeling*. Springer New York, 2010. doi: 10.1007/978-1-4419-0742-4. URL https://doi.org/10.1007/978-1-4419-0742-4.

Aryo Pradipta Gema, Joshua Ong Jun Leang, Giwon Hong, Alessio Devoto, Alberto Carlo Maria Mancino, Rohit Saxena, Xuanli He, Yu Zhao, Xiaotang Du, Mohammad Reza Ghasemi Madani, Claire Barale, Robert McHardy, Joshua Harris, Jean Kaddour, Emile van Krieken, and Pasquale Minervini. Are we done with MMLU? NAACL, 2025. URL https://doi.org/10.18653/v1/2025.naacl-long.262.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437. URL https://doi.org/10.1198/016214506000001437.

Isabelle Guyon, Gavin C. Cawley, Gideon Dror, and Vincent Lemaire. Results of the active learning challenge. In *JMLR Workshop and Conference Proceedings, Vol. 16 (Active Learning and Experimental Design)*, pages 19–45, 2011. URL https://proceedings.mlr.press/v16/guyon11a.html.

Alex Kipnis, Konstantinos Voudouris, Luca M. Schulze Buschoff, and Eric Schulz. metabench – a sparse benchmark of reasoning and knowledge in large language models. ICLR (per arXiv comment), 2025. URL https://arxiv.org/abs/2407.12844.

Yehuda Koren, Robert Bell, and Chris Volinsky. Matrix factorization techniques for recommender systems. *Computer*, 42(8):30–37, 2009. doi: 10.1109/MC.2009.263. URL https://doi.org/10.1109/MC.2009.263.

Jannik Kossen, Sebastian Farquhar, Yarin Gal, and Tom Rainforth. Active testing: Sample-efficient model evaluation. ICML (per arXiv comment), 2021. URL https://arxiv.org/abs/2103.05331.

John P. Lalor, Hao Wu, and Hong Yu. Building an evaluation scale using item response theory. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 648–657, 2016. doi: 10.18653/v1/D16-1062. URL https://doi.org/10.18653/v1/D16-1062.

F. M. Lord. *Applications of Item Response Theory to Practical Testing Problems*. Routledge, 2012. doi: 10.4324/9780203056615. URL https://doi.org/10.4324/9780203056615.

David J. C. MacKay. The evidence framework applied to classification networks. *Neural Computation*, 4(5):720–736, 1992. doi: 10.1162/neco.1992.4.5.720. URL https://doi.org/10.1162/neco.1992.4.5.720.

Lovish Madaan, Aaditya K. Singh, Rylan Schaeffer, Andrew Poulton, Sanmi Koyejo, Pontus Stenetorp, Sharan Narang, and Dieuwke Hupkes. Quantifying variance in evaluation benchmarks. arXiv preprint, 2024. URL https://arxiv.org/abs/2406.10229.

Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. tinyBenchmarks: Evaluating LLMs with fewer examples. ICML (PMLR 235; proceedings.mlr.press/v235/maia-polo24a.html), 2024. URL https://arxiv.org/abs/2402.14992.

Evan Miller. Adding error bars to evals: A statistical approach to language model evaluations. arXiv preprint, 2024. URL https://arxiv.org/abs/2411.00640.

Carl N. Morris. Parametric empirical Bayes inference: Theory and applications. *Journal of the American Statistical Association*, 78(381):47–55, 1983. doi: 10.1080/01621459.1983.10477920. URL https://doi.org/10.1080/01621459.1983.10477920.

Allan H. Murphy. A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4):595–600, 1973. doi: 10.1175/1520-0450(1973)012<0595:ANVPOT>2.0.CO;2. URL https://doi.org/10.1175/1520-0450(1973)012<0595:ANVPOT>2.0.CO;2.

Curtis G. Northcutt, Anish Athalye, and Jonas Mueller. Pervasive label errors in test sets destabilize machine learning benchmarks. NeurIPS Datasets and Benchmarks (per arXiv comment), 2021. URL https://arxiv.org/abs/2103.14749.

David Owen. How predictable is language model benchmark performance? arXiv preprint, 2024. URL https://arxiv.org/abs/2401.04757.

Lorenzo Pacchiardi, Lucy G. Cheke, and José Hernández-Orallo. 100 instances is all you need: Predicting the success of a new LLM on unseen data by testing on a few instances. arXiv preprint, 2024. URL https://arxiv.org/abs/2409.03563.

John C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In Alexander J. Smola, Peter L. Bartlett, Bernhard Schölkopf, and Dale Schuurmans, editors, *Advances in Large-Margin Classifiers*. The MIT Press, 2000. doi: 10.7551/mitpress/1113.001.0001. URL https://doi.org/10.7551/mitpress/1113.001.0001.

Yangjun Ruan, Chris J. Maddison, and Tatsunori Hashimoto. Observational scaling laws and the predictability of language model performance. NeurIPS (per arXiv comment), 2024. URL https://arxiv.org/abs/2405.10938.

Donald B. Rubin. Inference and missing data. *Biometrika*, 63(3):581–592, 1976. doi: 10.1093/biomet/63.3.581. URL https://doi.org/10.1093/biomet/63.3.581.

Christoph Sawade, Niels Landwehr, Steffen Bickel, and Tobias Scheffer. Active risk estimation. In *Proceedings of the 27th International Conference on Machine Learning (ICML)*, pages 951–958, 2010. URL https://icml.cc/Conferences/2010/papers/285.pdf.

Stanford AIMS. Measurement Data Bank (aims-foundations/measurement-db). Hugging Face dataset, revision 54074453 (2026-10-07); curation CC BY-SA 4.0, benchmarks under their upstream licences, 2026a. URL https://huggingface.co/datasets/aims-foundations/measurement-db.

Stanford AIMS. NeurIPS 2026 Predictive AI Evaluation Competition (PAIEC): Competition description and rules. Codabench competition page and starter kit, 2026b. URL https://aims.stanford.edu/competition.

Stanford AIMS. paiec_baseline: Starter kit and baselines for PAIEC 2026. GitHub repository aims-foundations/paiec_baseline, commit cc88a959 (2026-10-15), 2026c. URL https://github.com/aims-foundations/paiec_baseline.

Sang Truong, Yuheng Tu, Percy Liang, Bo Li, and Sanmi Koyejo. Reliable and efficient amortized model-based evaluation. ICML (PMLR 267; proceedings.mlr.press/v267/truong25c.html), 2025. URL https://arxiv.org/abs/2503.13335.

Rajan Vivek, Kawin Ethayarajh, Diyi Yang, and Douwe Kiela. Anchor points: Benchmarking models with much fewer examples. EACL, 2024. URL https://doi.org/10.18653/v1/2024.eacl-long.95.

David H. Wolpert. Stacked generalization. *Neural Networks*, 5(2):241–259, 1992. doi: 10.1016/S0893-6080(05)80023-1. URL https://doi.org/10.1016/S0893-6080(05)80023-1.

Lexin Zhou, Lorenzo Pacchiardi, Fernando Martínez-Plumed, Katherine M. Collins, Yael Moros-Daval, Seraphina Zhang, Qinlin Zhao, Yitian Huang, Luning Sun, Jonathan E. Prunty, Zongqian Li, Pablo Sánchez-García, Kexin Jiang Chen, Pablo A. M. Casares, Jiyun Zu, John Burden, Behzad Mehrbakhsh, David Stillwell, Manuel Cebrian, Jindong Wang, Peter Henderson, Sherry Tongshuang Wu, Patrick C. Kyllonen, Lucy Cheke, Xing Xie, and José Hernández-Orallo. General scales unlock AI evaluation with explanatory and predictive power. arXiv preprint, 2025. URL https://arxiv.org/abs/2503.06378.

A Derivations: decomposition, learning curve, bounds, protocol dependence, acquisition lemmas

Notation: pair (s, b) with N graded items; item i has n_i recorded responses with k_i successes, $p_i = k_i/n_i$. Per pair and seed the protocol permutes the items, takes the first $N_A = \lfloor N/2 \rfloor$ as acquisition half A and the rest as evaluation half E; budget B reveals the first B items of A with one Bernoulli(p_i) response each, $k = \sum y_i$; the score is $\mathrm{Brier}_B = \sum_{i \in E}(n_i q_i^2 - 2q_i k_i + k_i)/\sum_{i \in E} n_i$, which for a pair-constant q equals $(q - R_E)^2 + U_E$ with $R_E = \sum_E k_i/\sum_E n_i$ and $U_E = R_E(1 - R_E)$. Numbers marked "discovery" are Design A, seeds 0–2.

A.1 Exact per-pair decomposition. For any predictor returning q_i on E, $\mathrm{Brier}_B = (\bar{q} - R_E)^2 + R_E(1 - R_E) + G_B$ with $\bar{q}$ the response-weighted mean of the q_i and G_B the difference between the item-wise and the pair-constant score. *Proof.* Per item $(nq^2 - 2qk + k)/n = (q - p)^2 + p(1 - p)$; averaging the constant $\bar{q}$ over E with weights n_i gives $(\bar{q} - R_E)^2 + \mathrm{Var}_w(p) + \mathrm{mean}_w\, p(1 - p) = (\bar{q} - R_E)^2 + R_E(1 - R_E)$ by the identity $m(1 - m) = \mathrm{mean}\, p(1 - p) + \mathrm{var}\, p$ for the weighted mean m. G_B is non-zero only with item-level information. □ Consequences: $U = 0.18328$ on the discovery test pairs (2,142 pairs × 3 seeds); the submitted model's reliability term (binned, $S = 3$) is ≤ 0.0006 at every budget (0.00053 at $B = 0$, where binning violates the Murphy identity by 0.0002) and its resolution 0.0022 of $\mathrm{Var}(R_E) = 0.0667$ at $B = 0$; G_B is bounded below by the item-difficulty oracle.

A.2 Learning curve of a posterior-mean predictor. Fixed prior mean μ and strength κ, $q_B = (k + \kappa\mu)/(B + \kappa)$; $\bar{p}_A$ the unweighted mean of p_i over A, $\bar{p}$ over all items, $\delta = \bar{p}_A - R_E$. (1) $q_B - R_E = [(k - B\bar{p}_A) + \kappa(\mu - R_E) + B(\bar{p}_A - R_E)]/(B + \kappa)$: sampling error, prior error, split error. (2) Given the split, $\mathrm{E}[k \mid A] = B\bar{p}_A$ and $\mathrm{Var}[k \mid A] = B\,W_A + B\,V_A(N_A - B)/(N_A - 1)$ with W_A the mean of $p_i(1-p_i)$ and V_A the population variance of p_i over A (hypergeometric factor). (3) Cross terms with $(k - B\bar{p}_A)$ vanish. (4) Over a random split δ is symmetric, so $\mathrm{E}[(\mu - \bar{p})\delta] = 0$; with $R_E = \bar{p} - \delta/2$, $\mathrm{E}[(\mu - R_E)\delta] = D/2$, $D = \mathrm{E}\,\delta^2$. (5) $V_0 = \mathrm{E}(\mu - R_E)^2$ is the budget-0 excess, used as measured. (6) $(B + \kappa)^2\,\mathrm{E}(q_B - R_E)^2 = B\,(U - (B-1)F) + \kappa^2 V_0 + B(B + \kappa)D$, using $\mathrm{E}[W_A + V_A] = U$ and $F = \mathrm{E}[V_A/(N_A - 1)]$. (7) Hence Eq. 2. (8) For a random halving of N values with variance V, $\mathrm{Var}(\mathrm{mean}_A) = V/(N-1)$ and $\delta = 2(\mathrm{mean}_A - \bar{p})$, so $D = \mathrm{E}[4\,\mathrm{Var}_i(p_i)/(N-1)]$, a protocol quantity; $F \approx D/2$. ALC $= U + 0.1V_0 + \sum_{B \geq 1} w_B\,\mathrm{E}(q_B - R_E)^2$ splits into prior, sampling and split terms (discovery: 0.0117, 0.0107, 0.0008; B_0 0.0067; formula ALC 0.2132 against 0.2131 observed). *Disclosure.* The two-term form is the textbook MSE of a Beta posterior mean [Morris, 1983]; the split term was added on 2026-10-20 after a 13% shortfall at $B = 31$; its first version used $D \approx 2r(1-r)/N$ and a squared coefficient; the derivation above, the same day and before any confirmatory run, corrected the coefficient to $B/(B + \kappa)$, D to $4\,\mathrm{Var}_i(p_i)/(N-1)$ and added F. A Monte-Carlo test with the exact protocol mechanics (600 synthetic pairs $\times$ 60 replications) gives a summed relative error of 1.5% for the exact form and 27.6% for the earlier approximation. *Discovery comparison* ($U = 0.18328$, $V_0 = 0.06734$, $\kappa = 2.7268$, $D = 0.001382$, $F = 0.000695$), relative error of Eq. 2 against the observed beta-binomial $(q - R_E)^2$ at $B = 1, 3, 7, 15, 31$: $S = 1$: -0.1, $+0.1$, $+0.8$, $+0.4$, $+0.3$% (two-term: -0.8 to -11.2%); $S = 3$: -0.4 to -5.9%; $S = 30$: -0.5 to -8.3% (identity collision, A.3).

A.3 Validity limits (stated before the confirmatory runs). (1) Fixed prior strength and a label-independent prior mean: the submitted model violates this through pooling; with its own B_0 excess (0.06522) the formula overstates its $(q - R_E)^2$ at $B = 1$ by $+17.6$% ($S = 3$) and $+78.4$% ($S = 30$), vanishing by $B = 31$ ($+0.1$%, $+2.2$%); hence $V_0(S, B)$ and H2b. (2) Identity collision for $S \geq 2$. (3) Exchangeable items, one response per revealed item, equal halves; R_E is response-weighted while $\bar{p}_A$ is not (they coincide when n_i is constant within the pair, 69% of test pairs); second order in $1/N$. (4) In-sample moments: the test is of the budget dependence (five points), not an out-of-sample prediction. (5) $\mathrm{E}[R_A(1 - R_A)] = \mathrm{E}[R_E(1 - R_E)]$ by symmetry of the split.

A.4 What follows. $\kappa^* = U/V_0$ minimises every budget's error (discovery: $U/V_0 = 2.722$, $\bar{\kappa}_{\mathrm{ML}} = 2.731$, cluster-bootstrap SE 0.31; a moment identity for a well-specified Beta prior, so a specification test); with the split term $\kappa^* = 2.76$. The formula puts 77.5% of the excess over U at budgets 0–3 (observed baseline 77.2%, submitted model 78.5% at $S = 3$). Even with $V_0 = 0$ a text-free predictor cannot go below $U + 0.0107 + 0.0008 = 0.1948$ ALC. D and F scale with $1/N$: D is 0.0090 at $N = 80$ and 0.0007 at $N = 1{,}000$ ($V = 0.177$).

A.5 Bound ladder and L10. Rungs at $B = 0$ (Brier; discovery test pairs): L0 constant 0.5: 0.2500; L1 submitted model: 0.2485; L3′ ability only (fold-honest θ): 0.2480; L3″ ability only (in-sample θ): 0.2460; L2b benchmark location on the logit scale (in-sample ALS): 0.2211; L2 leave-one-out benchmark location (mean rate of the other pairs): 0.2067 (65% of $\mathrm{Var}(R)$ explained); pooling oracle (Beta posterior around the leave-one-out mean of the other test pairs of the benchmark, strength 8 chosen in-sample, $S = 1$): 0.2069; L4′/L5′ location plus ability: 0.2065/0.2055; L6 full in-sample ALS (unrealistic): 0.1885; L7 pair-rate oracle: 0.1833 ($= U$, ALC floor of pair-constant predictors); L8 rate plus leave-one-out item difficulty: 0.1233 (-0.054 ALC against the pair-level model, -0.048 already at $B = 0$); L9 oracle k/n over all cells: 0.0065 (biased: cells with $n = 1$ contribute 0; the value 0.0229 in the repeat statistics is a different estimand, response-weighted over the whole database). *L10 estimator:* for a cell with $n \geq 2$ responses, $\mathrm{E}[k(n-k)] = n(n-1)p(1-p)$, so $k(n-k)/(n(n-1))$ is unbiased for $p(1-p)$. Weighting cells by n over the 3.14 M cells with $n \geq 2$ (10.0% of cells, 58.6% of all 68.3 M responses) gives 0.0444; over the 62 benchmarks with ≥ 100 such cells the median is 0.0488 (IQR 0.0106–0.0881); a second computation with different code (pandas on the raw responses, `var(ddof=1)`) reproduces the median exactly (all summary values within 5×10^{-16} relative). Caveats: repeated responses exist only where curators recorded trials; the answer-weighted mean is dominated by one benchmark (monkey_power_laws, 72% of the repeated responses; without it 0.068); some benchmarks show zero noise at $n = 2$ (greedy decoding or de-duplication, a conjecture).

A.6 Protocol dependence. Subjects per benchmark: gain against the baseline $-0.0003 \rightarrow -0.0030 \rightarrow -0.0047 \rightarrow -0.0130$ ALC from one to thirty (discovery; in the formula $V_0(S, B)$). Split per pair versus per item: G_B exists only if evaluation items carry foreign labels. Items per pair: $D, F \propto 1/N$. Items with foreign labels: G_B linear in their number up to about 50 items ($S = 3$: -0.00009 per item at $B = 31$, R^2 0.995), saturating beyond ($S = 30$ classes 21–50 / 51–100 / > 100: -0.0157 / -0.0125 / -0.0138 discovery; -0.0171 / -0.0123 / -0.0140 confirmatory, H8 passed). Indistinguishable subjects: surplus labels for about 2% of rows, shifting the baseline curve between regimes by up to 8.3% at $B = 31$.

A.7 Acquisition lemmas. The j-th acquired label is contained in all budgets $B \geq j$, so its cumulative ALC weight is $W(j) = 0.9$ ($j = 1$), 0.7 ($j = 2$–3), 0.5 (4–7), 0.3 (8–15), 0.1 (16–31); labels 1–7 carry 52% of the total weight, 1–15 carry 81%. *Lemma 1 (fallback to random).* If no candidate item carries labels of other subjects, all candidates have identical information value, so any budget-respecting policy equals random acquisition in expectation; a policy built on the platform's sha256-uniform with the same threshold is bit-identical to the random policy (checked mechanically: 0 differing decisions on 15,000 candidates). *Lemma 2 (ignorability).* If the purchase decision depends only on quantities conditionally independent of the pair's own outcome given the pool (presence of foreign labels, context, the shared uniform), the acquired set is a random subset of the pool with respect to the outcomes and the (k, n) likelihood for R stays unbiased [Rubin, 1976]; a decision that depends on an estimate of item difficulty is ignorable only if the likelihood conditions on it, otherwise biased as for uncertainty sampling with the Empirical Mean [Kossen et al., 2021, Sawade et al., 2010]. Reproduction: `python3 -I theorie/alc_law.py --U 0.18328 --V0 0.06734 --kappa 2.7268 --D 0.001382 --F 0.000695`; tests `test_alc_law.py` (11 tests, including one deliberately triggered failure).

B Pre-registration: frozen rules

The plan `prereg/PRAEREGISTRIERUNG-v2.md` (repository path) was frozen at 2026-10-20 20:26:04 UTC; sha256 `e5fd821fdf6e2db3be86e1e341b02bc4199d4721790dd3ae8f027b60e9521ab7`; 24,338 bytes, 181 lines. At freezing no confirmatory run existed (the confirmation folder did not exist; the hash file records the listing). The file is never edited; amendments would be separate dated files with their own hash. Amendment log at the time of this report: none. The frozen text (English) states:

Scope and honesty frame. All hypotheses arose from one data release (156 benchmarks, 142 with usable pairs, 12,918 pairs) on the discovery randomness (fold seed 1234, seeds 0–2) and are tested on fresh randomness only (fold seed 200, seeds 200–202); no new benchmarks exist. Model under test: the submitted predictor, package v2.0-core (mathematically v1 plus hardening, bit-identical to v1 on the discovery sets, with the single documented exception that a nested Platt stage selection may change the stage set $\{0, 1, 3, 7\}$). Rule R1: if the submitted model changes, every confirmatory run is repeated for the new model. *Deviations, recorded 2026-10-21:* (1) the exception occurred: on the discovery fold assignment the nested selection kept stages 0 and 1 in every fold, stage 3 in 8 and stage 7 in 7 of 10 folds, so the submitted package carries stages $\{0, 1\}$ (intersection rule fixed in advance); the confirmatory runs used, per fold, that fold's own nested stage set (Design A: $\{0, 1, 3, 7\}$ in all ten folds, stage 15 in five and 31 in two; Design B: $\{0, 1, 3\}$ in all, stage 7 in five, 15 in two) rather than the package's exact set; sensitivity runs with the v1 set $\{0, 1, 3, 7\}$ and with the v1 pooled choice gave the same verdict in all 23 comparison rows (largest shift 2.4×10^{-5} ALC); no run was repeated with exactly $\{0, 1\}$. (2) The discovery numbers, the league and Appendix D were computed along the v1 path (stages $\{0, 1, 3, 7\}$); on 1,079,568 phase-1 records the two paths are bit-identical, in phase 2 they differ by $+2 \times 10^{-5}$ to $+7 \times 10^{-5}$ pair-level ALC. (3) H2c: the frozen symbol $\bar{\kappa}_{\mathrm{ML}}$ denotes the fold-wise $\alpha + \beta$ of the beta-binomial prior, which the code fits by the method of moments, not by marginal likelihood; the computation is the one used in discovery. Protocol fact stated in advance: identity collision (Section 2), so $S = 1$ is the clean regime for H2a.

Estimand and decision rule. As in Section 5, including the operating characteristics table: $P(\text{equivalent} \mid \text{null})$ is 100%, 99.7%, 82%, 38% and 0% for SE 0.0001, 0.0002, 0.0003, 0.0004 and ≥ 0.0005; MDE (80% power) 0.0003, 0.0006, 0.0009, 0.0011, ≥ 0.0014. No comparison with SE ≥ 0.0005 is declared equivalent. About 20 verdicts, no multiplicity correction, expectation of ≈ 0.5

false directional verdicts stated (this counted one direction only; the two-sided expectation is ≈ 0.9, Section 5).

Learning-curve formula. Eq. 2 with fixed estimation recipes for U, V_0, κ (pair-weighted mean of the fold-wise κ_{ML}), D and F (from the item tables of the test pairs); the earlier approximation and the two-term form are sensitivity rows only.

Hypotheses (frozen wording, with the clause for failure). H1 the submitted model beats the beta-binomial baseline when ≥ 2 subjects share `labeled`, not when one does (fails: the pooling claim is dropped for the failing regime). H2a the formula matches the baseline curve within $\pm 10\%$ at all five budgets in Design A $S = 1$ and Design B $S = 1$; expectation $\leq 3\%$ (fails: reported as an approximation within $X\%$; Section 3 keeps the decomposition and the ladder). H2b pooling acts as an S- and budget-dependent reduction of the prior error: six adjacent-regime steps at $B = 1, 3$ all better; fixed-κ deviation at $S3/B1$ in $[+10, +30]\%$ and at $S30/B1$ in $[+50, +110]\%$. H2c $|\bar{\kappa}_{\mathrm{ML}} - U/V_0| \leq 2\,\mathrm{SE}_{\mathrm{boot}}$ (fails: the Beta prior is reported as misspecified in the ALC-relevant moments). H3 $\geq 70\%$ of the excess over the pair floor at budgets 0–3 in $S = 1, 2, 3$ (lower CI bound). H4 subject identity has no relevant effect: equivalent in $S = 1, 2, 3$ (fails if better: reported as a small real effect with the number). H5 convex mixture, logit stacking and NPEB prior: nine comparisons all equivalent, none better. H6 leave-one-benchmark-out $R^2 \leq 0.05$ of ridge on fixed feature set B for R_b and logit R_b. H7 L10 median ≥ 0.02 over benchmarks with ≥ 100 repeated cells; second computation within 1%. H8 item-stage gain linear in foreign-labelled evaluation items up to ≈ 50, saturating beyond. **League rule.** Protocol identity (max $|\Delta q| = 0$ for EM and BB) is a precondition; inner 75/25 tuning per fold, seed 100; verdicts in both directions; a family without documented tuning is "not evaluated", never "beaten"; time cap 2 h per family. **Gates** for any model change (stratum stage, text prior, acquisition function) with rules only and at most one change; none passed or was evaluated in time, so the unchanged predictor is final (Appendix H). **Reporting commitments.** One JSON object per hypothesis with code-produced verdicts, fail-closed to "not evaluated"; every Table 2 number points to a field of that file; discovery values stay visible; nothing is re-tuned after the confirmatory run.

C Protocol replica, tests, hardening, runtime

Replica. Per seed, `modell/simulator.py` builds the 50/50 split of every test pair (random permutation, first half acquisition), the acquisition order (first B of a random permutation of the acquisition half; the platform's selection sampling with acceptance probability *labels remaining / items remaining* yields a uniformly random subset of the same size), one recorded response per revealed item, and the exact Brier score over the evaluation items. Subjects are presented through the eight visible fields only; items through unique placeholders; benchmarks through an anonymous code. Each run receives one `labeled` set per budget; the budget-0 call starts from an empty cache.

A leak found and fixed. An early version reused the grouping cache of the previous run at budget 0, so the model saw labels of another run of the same benchmark; it was noticed because the model without subject knowledge was better at budget 0 than any constant (0.2239 against 0.2485). All results before the fix were discarded; a regression test fails against the old code and passes against the new one.

Assumptions. (A1) a run contains one benchmark (on the platform `labeled` also holds other benchmarks, filtered by `benchmark_id`); (A2) the cap of 30 pairs per benchmark weights benchmarks more evenly than uniform pair drawing; (A3) the split is drawn per pair, as the competition text reads; a per-benchmark split would silence the item stage and the submission falls back to its pair-level results; (A4) the audit exclusions also apply to the test pairs of the replica, for model and references alike; (A5) `test_condition` is pooled. The item field `interactors` varies within (subject, item) in none of the 142 benchmarks.

Tests and hardening. 26 unit tests (Laplace mode and inverse Hessian against `scipy`; budget stages; the competition's illustrative ALC example; the Brier identity; two leak regressions; the item stage; cached path equal to uncached; sampled cache key equal to the exact key; label parsing; fallback chain). The submitted package (v2.0-core) triggered seven deliberate failure cases (broken state, unreadable labels, exceptions in the hierarchical path, thread contention, oversized label sets, empty inputs, missing `benchmark_id`), each first red then green, and produced 0 fallbacks on 10^4 competition-shaped inputs and 10^4 fallbacks on a deliberately broken state; relative to the v1 package

its predictions are bit-identical on 1,079,568 phase-1 records (max $|\Delta q| = 0$) and on 1,214,136 phase-2 records except in the documented Platt stages 3 and 7 (max $|\Delta q| = 0.025$). Both package variants are in the repository; the submitted file is `paiec_v2core_final.zip`.

Runtime. Per call with 100,000 labels passed as a fresh copy: median 0.38 ms; first call 0.27 s; with 300,000 labels 0.45 ms; memory increase 54 MB RSS at 300,000 labels. Grouping is cached per label set; up to 1,024 labels the key is a hash over all entries, above that length plus 64 sampled entries, sound under the published protocol where `labeled` grows between checkpoints and is identical for all targets of a checkpoint. If the platform started a new process per call or copied `labeled` per call, the first-call cost would recur; this would equally affect the organisers' baselines and cannot be tested from outside.

D Full tables and figures

Table 4: Protocol sensitivity: what moves the error besides the model. Discovery (Design A, seeds 0–2) unless stated; paired differences ± cluster-robust SE. Formula rows use the three-term learning curve with moments from the S1 test pairs ($U = 0.1833$, $V_0 = 0.0673$, $\kappa = 2.73$, $D = 0.00138$, $F = 0.00070$; nothing fitted). Sources: P2 = `modell/ergebnisse/cv_A_p2_S1-S2-S3-S30-kompakt10/auswertung.json`; league = `liga/ergebnisse/liga_vergleich.json` (BB ALC, full value rounded once); D_exakt = `daten/D_exakt.json`; obs. = `daten/beobachtung_discovery.json`; alc_law = `theorie/alc_law.py`; loss = `daten/verlust.json`; P5 pilot = `daten/akquise_pilot_tabelle.json`; `daten/` = `report/src/daten/`.

protocol quantity	regime	effect (ALC unless stated)	reading	src
subjects per benchmark	S1	BB ALC 0.2131; v1 − BB -0.0003 ± 0.0004	one subject: nothing to pool; v1 gains at B0–B1 only; BB unchanged	P2, league
subjects per benchmark	S2	BB ALC 0.2133; v1 − BB -0.0030 ± 0.0006	pooling gain mainly at B1–B7; BB unchanged	P2, league
subjects per benchmark	S3	BB ALC 0.2134; v1 − BB -0.0047 ± 0.0007	pooling gain mainly at B1–B7; BB unchanged	P2, league
subjects per benchmark	S30	BB ALC 0.2135; v1 − BB -0.0130 ± 0.0013	pooling gain mainly at B1–B7; BB unchanged	P2, league
item overlap (foreign labels on evaluation items)	S3	item-stage gain -0.0004 ± 0.0001	grows with overlap; compact10 is hypothetical	P2
item overlap (foreign labels on evaluation items)	S30	item-stage gain -0.0037 ± 0.0006	grows with overlap; compact10 is hypothetical	P2
item overlap (foreign labels on evaluation items)	compact10	item-stage gain -0.0115 ± 0.0010	grows with overlap; compact10 is hypothetical	P2
50/50 item split (split term D)	all	$D = 0.00138$ (exact), 0.00072 (first approximation); ALC contribution of the split term 0.00078	protocol quantity, not model error	D_exakt, alc_law
items per pair N (finite-population term F)	all	$F = 0.00070$; N median 1000, min 80	second order in $1/N$	D_exakt
identity collision (surplus labels from indistinguishable subjects of other pairs)	S1 / S3 / S30	formula error for BB at B31: $+0.3\,\%$ / $-5.9\,\%$ / $-8.3\,\%$	clean only in S1; grows with S	obs., alc_law
subject known in training (name) vs unknown	S3, B0 / B3	excess known 0.0652 / 0.0239; unknown 0.0652 / 0.0276	B0 identical; B3 gap mostly benchmark composition	loss
label acquisition policy (coverage vs random), pilot	S3	$+0.00003 \pm 0.00012$ equivalent (260 pairs, 15 clusters)	pilot only (fold 0); full run not evaluated (aborted)	P5 pilot
label acquisition policy (coverage vs random), pilot	compact10	-0.00178 ± 0.00063 better (115 pairs, 5 clusters)	pilot only (fold 0); full run not evaluated (aborted)	P5 pilot
split per item instead of per pair	—	not evaluated (scenario aborted with the acquisition full run)	—	P5 pilot

Table 5: Ablations, discovery (Design A, seeds 0–2): paired ALC difference ± cluster-robust SE; letter = verdict by the pre-stated rule (b better, w worse, e equivalent, g grey zone; Section 5). The confirmatory H4 row (seeds 200–202) failed the equivalence hypothesis. Rows with a difference or SE below 0.00005 are given to five decimals. Sources: `modell/ergebnisse/cv_A_S1-S2-S3-S30/auswertung.json` (v1 stage set: Platt calibration, subject identity, benchmark slope), `modell/ergebnisse/cv_A_p2_S1-S2-S3-S30-kompakt10/auswertung.json` (item stage and item temperature), `konfirmation/hypothesen_tafel.json` (H4).

component (paired difference)	S1	S2	S3	S30
item stage (ours − pair model)	0.00000 ± 0.00000 e	-0.00022 ± 0.00004 b	-0.00045 ± 0.00008 b	-0.00366 ± 0.00061 b
Platt calibration per budget, v1 stage set (calibrated − raw)	-0.0009 ± 0.0006 g	-0.0007 ± 0.0006 g	-0.0006 ± 0.0005 g	-0.0003 ± 0.0004 g
subject identity (raw − without subject)	-0.0005 ± 0.0004 g	-0.0005 ± 0.0003 g	-0.0005 ± 0.0003 g	-0.0005 ± 0.0003 g
subject identity, calibrated (cal. − cal. without subject)	-0.0014 ± 0.0007 g	-0.0012 ± 0.0007 g	-0.0011 ± 0.0006 g	-0.0008 ± 0.0005 g
benchmark slope a_b (raw − without slope)	—	—	-0.00003 ± 0.00003 e	—
item temperature 1 vs 0.5 (ours − t0.5)	0.00000 ± 0.00000 e	-0.00005 ± 0.00001 b	-0.00009 ± 0.00002 b	-0.00064 ± 0.00012 b
item temperature 1 vs 2 (ours − t2)	0.00000 ± 0.00000 e	$+0.00002 \pm 0.00001$ w	$+0.00004 \pm 0.00002$ w	$+0.00025 \pm 0.00008$ w
subject identity, **confirmatory** Design A (H4, seeds 200–202)	-0.0005 ± 0.0003 g	-0.0006 ± 0.0003 b	-0.0007 ± 0.0003 b	—
stratum stage (item-metadata strata)	not evaluated: pilot on one fold only, full run aborted; not part of the submission			
text-based item prior	not evaluated: prepared, no measurement; not part of the submission			

Table 6: Brier score per label budget, discovery set (Design A, seeds 0–2; same runs as Table 1). Source: `modell/ergebnisse/cv_A_p2_S1-S2-S3-S30-kompakt10/auswertung.json`, `tabelle_alle[S][pred].brier_je_budget` (five decimals, cross-check only); cells are the full values in `liga/ergebnisse/liga_vergleich.json` (`regime[S].familien[pred].niveau[B`b`].mittel`), rounded once here.

Regime	Predictor	B_0	B_1	B_3	B_7	B_{15}	B_{31}
$S = 1$	ours	0.2485	0.2321	0.2160	0.2028	0.1945	0.1895
	BB	0.2506	0.2329	0.2159	0.2027	0.1943	0.1894
	EM	0.2500	0.3676	0.2459	0.2101	0.1961	0.1899
$S = 2$	ours	0.2485	0.2277	0.2119	0.2003	0.1933	0.1885
	BB	0.2506	0.2331	0.2160	0.2028	0.1945	0.1896
	EM	0.2500	0.3658	0.2454	0.2101	0.1963	0.1901
$S = 3$	ours	0.2485	0.2245	0.2092	0.1991	0.1926	0.1876
	BB	0.2506	0.2331	0.2161	0.2030	0.1947	0.1898
	EM	0.2500	0.3651	0.2455	0.2102	0.1965	0.1903

Table 7: Test-time hyperparameters per fold of Design A (discovery) and of the submitted state (trained on all benchmarks). μ_c, τ_c^2 and λ, τ_λ^2: mean and variance of the priors of c_b and a_b (Section 4); the pair deviation v_{sb} has variance $s_s^2 = \tau_v^2 + \kappa V_s$ for a subject whose name is known from training (V_s = uncertainty of its ability m_s; this κ is the coefficient of that term, unrelated to the prior strength κ of Eq. 2) and s_{unk}^2 for an unknown name; τ_i^2 = variance of the item offset; names = number of subject names with an ability estimate in the state. τ_λ^2 and κ vary strongly between folds, i.e. they are weakly identified. Sources: `daten/hyperparameter_je_falte_roh.json` (full-precision `hyp` entries of the ten fold states, rounded once here), `modell/paket/state_v2core.json` (submitted row).

Fold	μ_c	λ	τ_c^2	τ_λ^2	τ_v^2	κ	s_{unk}^2	τ_i^2	names
0	0.323	0.304	1.367	0.065	0.415	0.855	0.638	2.546	605
1	0.332	0.339	1.358	0.571	0.241	7.093	0.799	2.730	563
2	0.250	0.487	1.258	0.309	0.383	4.028	0.745	2.681	628
3	0.212	0.510	1.351	0.444	0.298	7.261	0.818	2.561	641
4	0.310	0.471	1.237	0.268	0.329	3.662	0.778	2.551	646
5	0.347	0.385	1.392	0.220	0.389	2.689	0.803	2.558	624
6	0.267	0.510	1.420	0.000	0.415	1.684	0.881	2.532	639
7	0.336	0.522	1.256	0.328	0.378	1.876	0.767	2.581	637
8	0.120	0.336	1.383	0.000	0.315	1.690	0.773	2.604	566
9	0.320	0.421	1.323	0.309	0.289	0.304	0.858	2.796	626
submitted	0.315	0.399	1.368	0.164	0.318	3.217	0.790	2.551	647

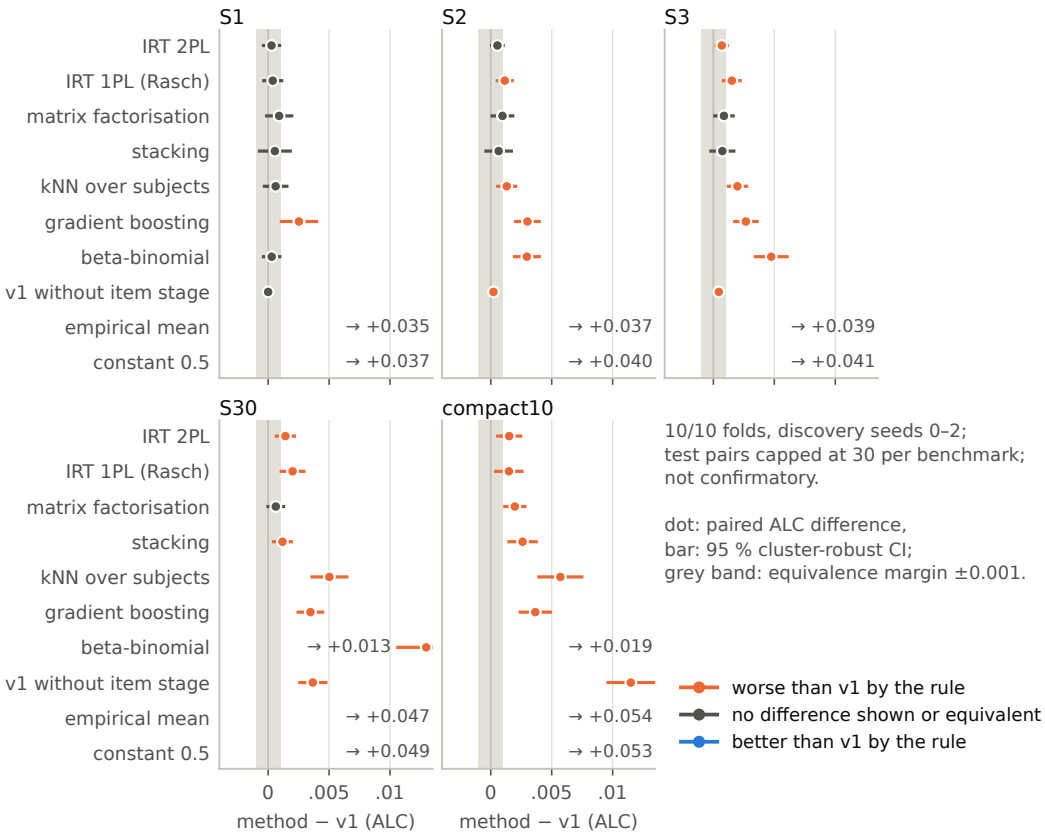


Figure 2: Method league (10 of 10 folds, discovery seeds): paired ALC difference method − submitted model ± 95% CI per regime ("v1" on the axis = the submitted predictor along the v1 path, Appendix B), equivalence band ±0.001. Sources: Table 10.

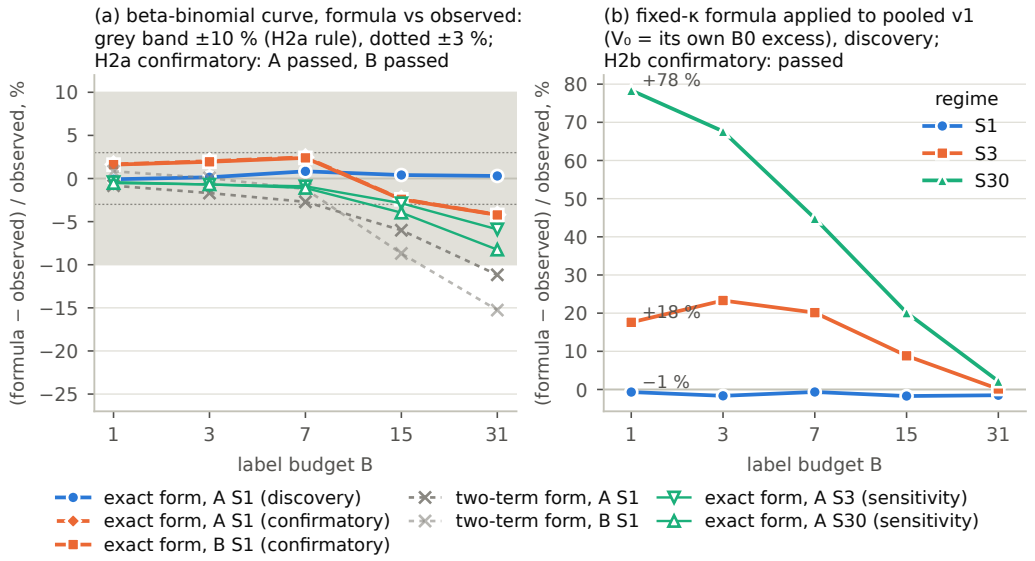


Figure 3: Left: relative error of Eq. 2 against the observed BB curve at $S = 1$, discovery (Design A) and confirmatory (Designs A and B), $\pm 10\%$ band; $S = 3$, $S = 30$ as sensitivity. Right: deviation of the fixed-κ form from the pooled submitted model (discovery; H2b). Sources: Table 10.

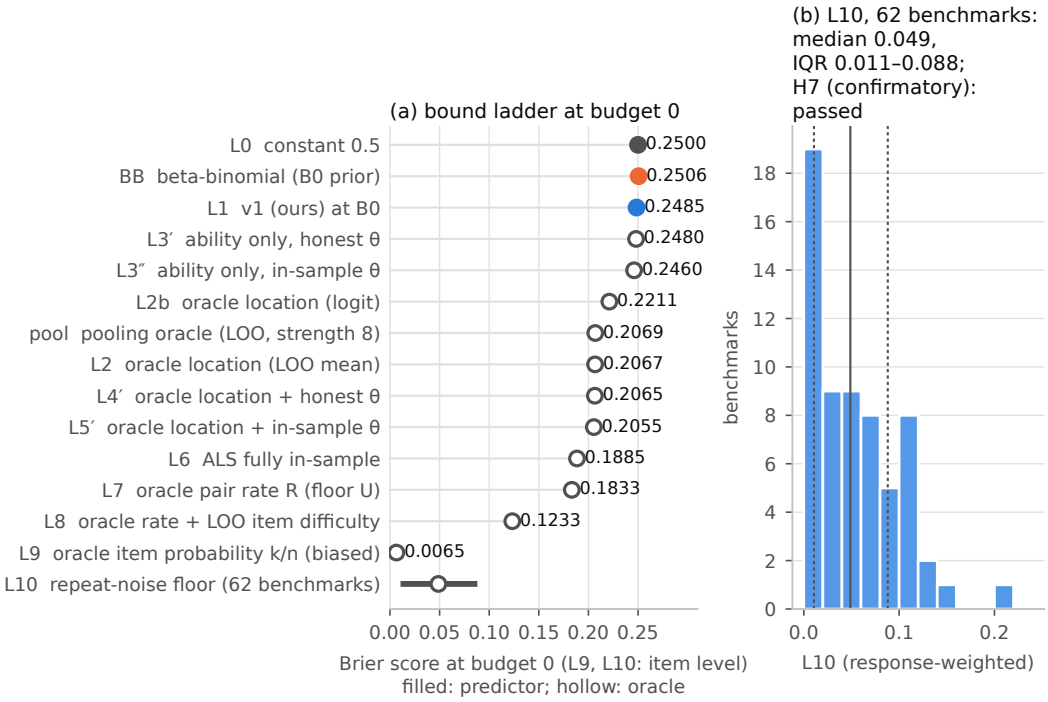


Figure 4: Bound ladder at budget 0 (Brier of oracles and predictors, discovery test pairs; Section 3, A.5) and the per-benchmark noise floor L10 over the 62 benchmarks with ≥ 100 repeated cells. Filled markers: predictors, hollow: oracles; the L9 and L10 rows are item-level. Sources: Table 10.

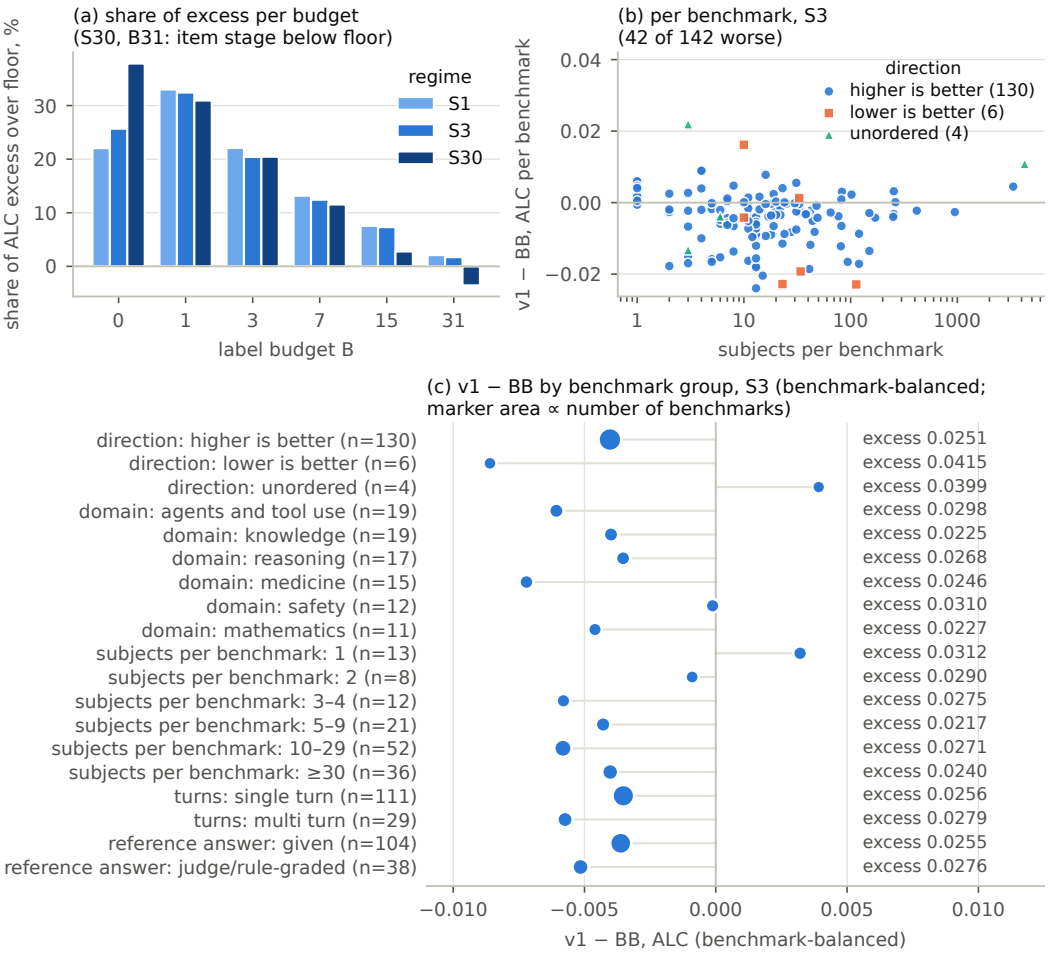


Figure 5: Error anatomy (discovery; (a) $S = 1, 3, 30$; (b), (c) $S = 3$). (a) Share of the excess over the pair-level floor per budget; (b) submitted $-$ BB per benchmark against the number of subjects in the benchmark (42 of 142 benchmarks worse than BB); (c) group means by polarity, domain, subjects per benchmark, turn type, reference answers (six largest domains; right: mean excess over the pair-level floor; polarity and turn type are missing for 2 of the 142 benchmarks).

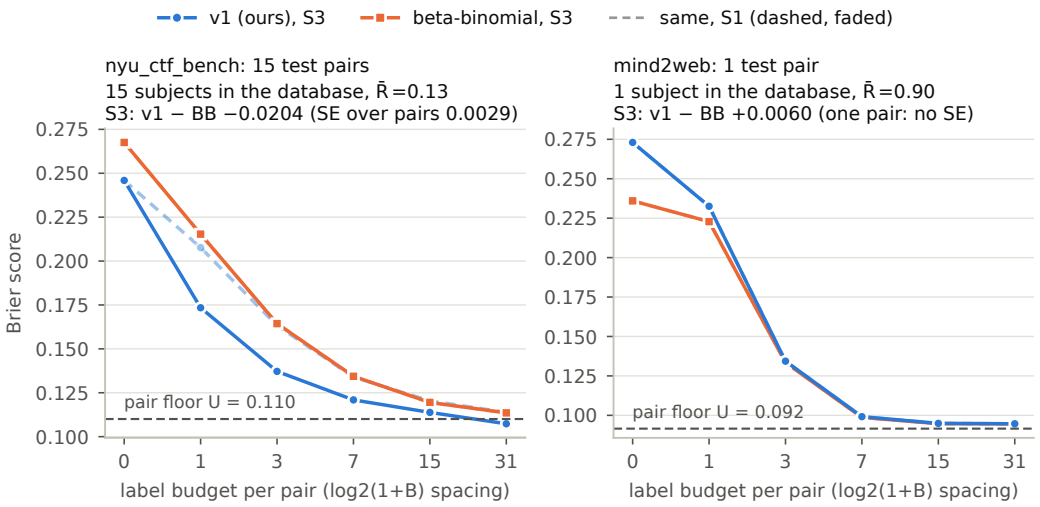


Figure 6: Two cases (discovery): NYU CTF Bench with one and three subjects sharing `labeled` (success), and Mind2Web, a single-subject benchmark (failure). Sources: Table 10.

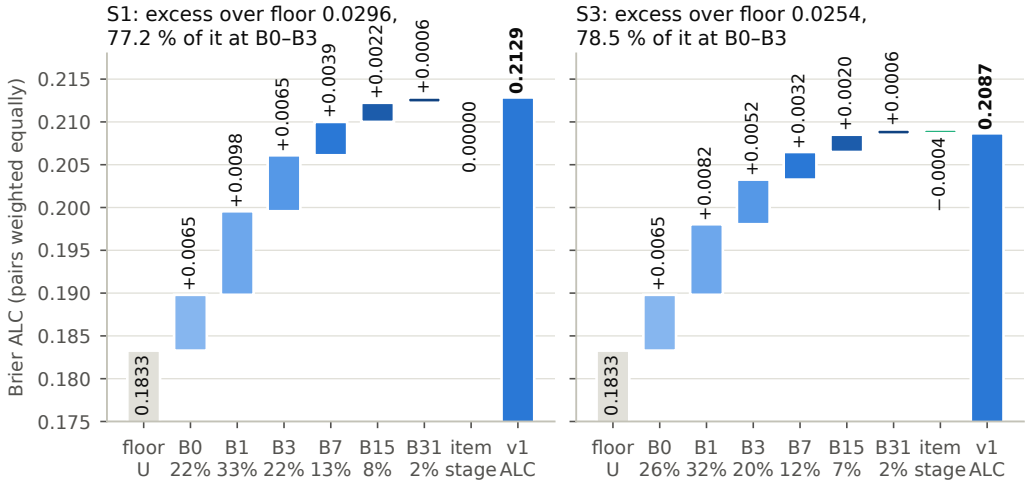


Figure 7: Decomposition of the ALC into pair floor and excess, per budget, $S = 1$ and $S = 3$ (discovery): floor U, ALC-weighted $(q - R)^2$ per budget (bar labels under the axis: share of the excess), item-stage gain, resulting ALC. Sources: Table 10.

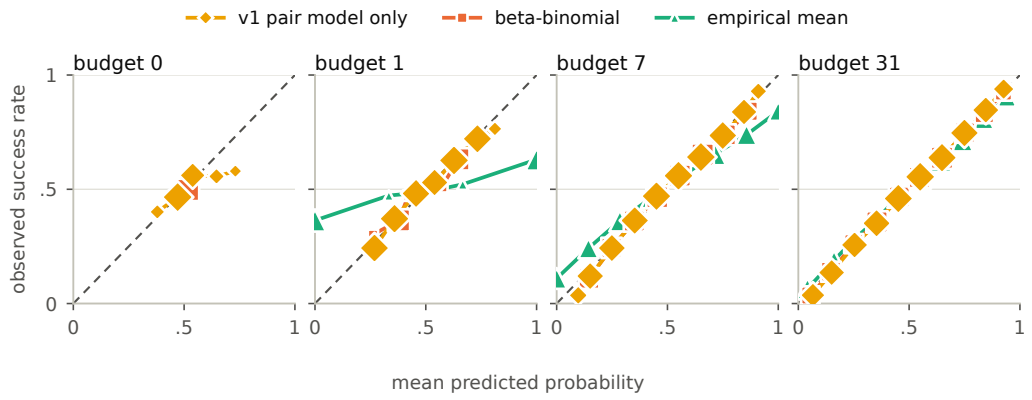


Figure 8: Reliability diagrams at budgets 0, 1, 7 and 31 ($S = 3$, discovery) for the pair-level model, BB and EM; bins with fewer than 20 pair records hidden; marker area $\propto \sqrt{n}$. Item-stage reliability: not evaluated (no item-level reliability table in the result set). Sources: Table 10.

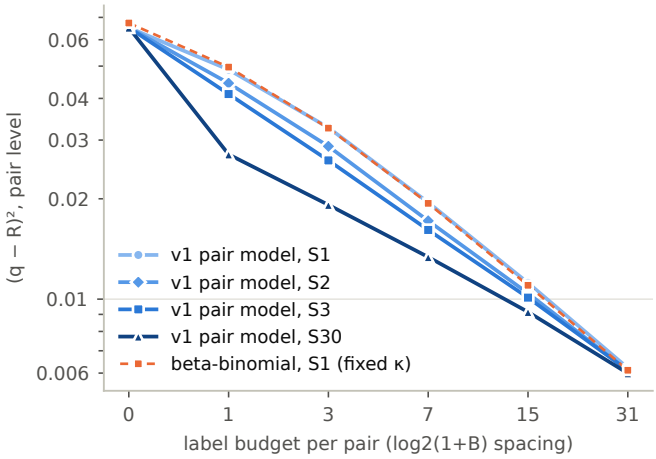


Figure 9: Effective prior error $V_0(S, B)$ of the pooled pair-level model per regime and budget (discovery): the quantity that replaces the fixed V_0 of Eq. 2 for the submitted model. Sources: Table 10.

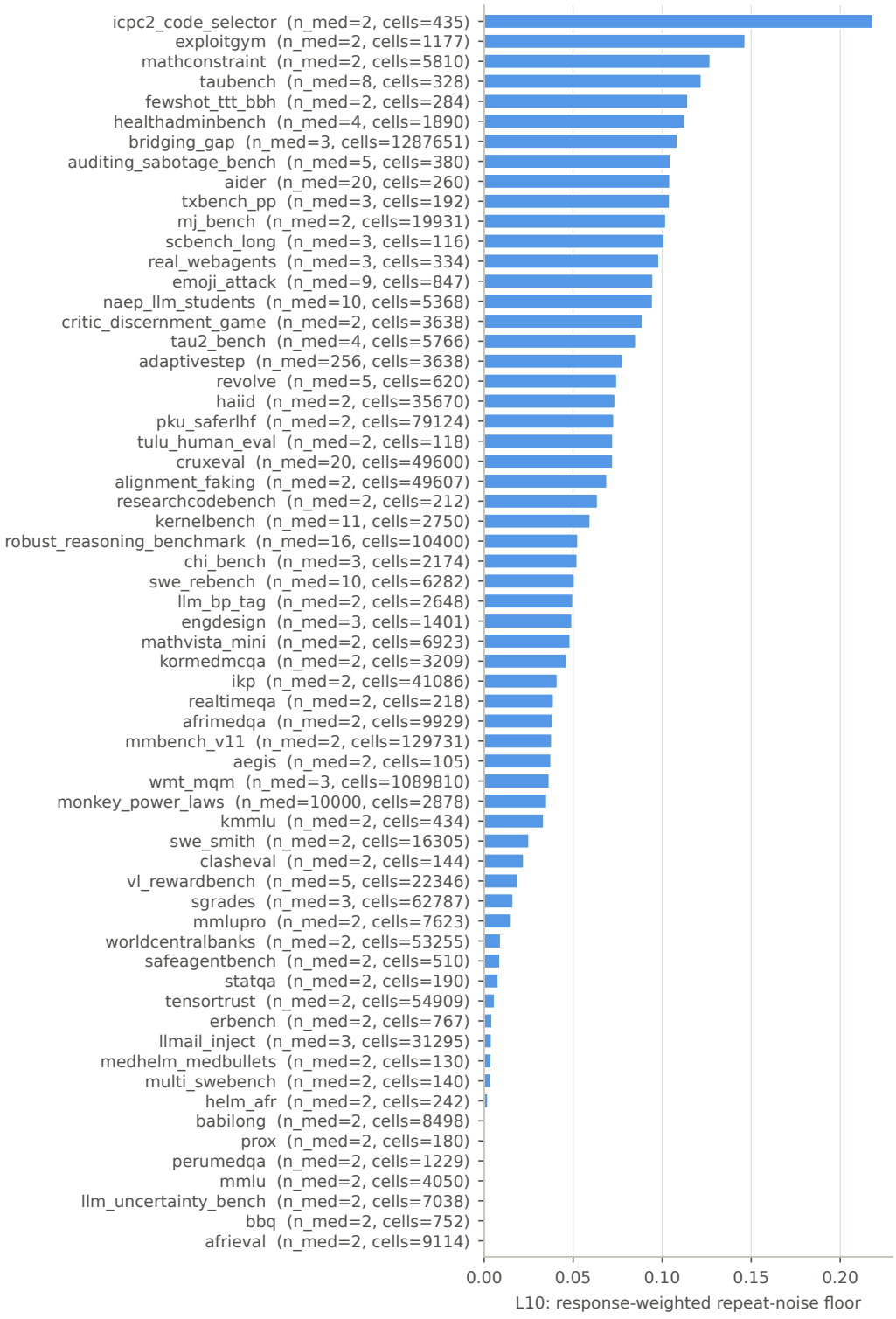


Figure 10: Response-noise floor L10 per benchmark (62 benchmarks with ≥ 100 repeated cells), the curation metric released as `noise_floor_by_benchmark.csv`; measurement-db revision `54074453`, second computation path (pandas) identical to the first (median exact; Table 9). n_med = median responses per repeated cell; cells = cells with $n \geq 2$. Sources: Table 10.

E Data-quality audit: findings, lists, evidence packs

Standard of evidence. An observation is included only if (i) its key numbers were obtained by at least two implementations with different code and, where possible, different data paths (Parquet footer statistics against the column itself, Hugging Face originals (SHA-256 checked against the LFS object id) against cached copies, upstream files fetched separately); (ii) where one exists, an external witness supports it (upstream file, publisher aggregate or paper); (iii) harmless explanations (documented intent, another reading of the columns) were checked; (iv) two further checks tried to refute it and failed. Where intent is plausible the observation is a question. Prior reports were searched in the Hugging Face discussions, measurement-db pull requests #1–#7 and paiec_baseline issues #1–#3 (as of 2026-10-20); none covered these points except an overlap with one baseline issue on stale embeddings. Everything was read-only; the proposed patches were checked with `git apply --check`, by re-implementing their selection logic and, in the evidence packs, by regression tests that fail without the patch and pass with it.

Table 8: The nine findings of the data-quality report, with the effect in the report's own wording. Revision `54074453`; checks run 2026-10-20. Source: Summary of the data-quality report sent to the organisers (not part of the repository; available on request); the public identifier-level lists are in `audit/`.

#	Finding	Scope	Stated effect (report wording)
1	`tools/prepare_data.py` sees 60 of 156 eligible graded benchmarks (`repo_info().siblings` stops at 100,000 files); BLE `get_benchmark` raises `KeyError`; embedding vectors keyed to pre-schema-3 item IDs	baseline payload; 96 benchmarks missing	BLE and BLE-acquisition baselines run on a truncated corpus; scoring itself not affected
2	proofaug: an upstream archive is public although the curation record says it remains local and the upstream README asks not to distribute it (question)	1 file, 68,666 bytes	none on measurements; concerns a third party's request
3	bird_sql: 64 of 120 configurations graded 0 on all 360 questions; publisher's aggregate 5.6–40.6%	23,040 grades (53% of the benchmark)	wrong ability signal for these models in the training data
4	swebench_live: 12 website-report subjects carry only successes (840 ones, no zeros); publisher rates 7–36% (question)	16 subjects (the 12 and those merged via website aliases)	20% of graded items carry only 1s; question whether the rule also applies in the test pool
5	Reasoning configuration not passed to the builder: configurations share one subject_id; in ikp stored as trial 1 and 2	ikp: 30 subjects, 42,140 rows	$< 1\%$ of eligible pairs; contradicts the subject definition and the meaning of `trial`
6	icpc2_code_selector: retrieval-free baseline run stored as trial 2 of gpt-4o on items showing retrieved candidates	435 rows	small; wrong stimulus for these rows
7	Winogrande twins collapsed in published human translations: identical prompts, opposite gold answers (bridging_gap, helm_afr)	117 upstream twin pairs (172 + 92 pairs in the database); 830 items	label noise in training data
8	119 different models in 13 benchmarks receive identical competition input (`normalized_name` null, label in no input field)	25 groups	in devbench all subject information is lost
9	Documentation: card frozen at 255 folders; licence-field texts; 127 stale manifests; inventory rule wording; `test_condition` contents; harness spellings; cross-benchmark item families	—	mostly none; two questions concern the test pool

Lists and evidence packs (folder `audit/` of the repository: identifiers, hashes and counts only; the packs with record excerpts went to the organisers). Lists F1–F9g (row counts of the full lists sent to the organisers; the public `audit/csv/` omits F2, F3b and F9b, and F1, F9c, F9d and F9e each omit one row for a benchmark folder under a separate note, see `audit/README.md`):
`F1_baseline_payload_coverage.csv` (281 folders), `F2_proofaug_archive.csv` (1), `F3a_bird_sql_configurations.csv` (120), `F3b_bird_sql_byte_identical_sql_graded_0.csv` (160), `F4_swebench_live_subjects.csv` (65), `F5a_ikp_merged_configurations.csv` (30), `F5b_reasoning_setting_only_in_label.csv` (218), `F6_icpc2_gpt4o_trials.csv` (435), `F7a`/`F7b` collapsed pairs (172, 92), `F8_identical_input_subject_groups.csv` (119), `F9a`–`F9g` (26, 9, 127, 278, 176, 14, 45 rows). Evidence packs (17): bird_sql, swebench_live, ikp, reasoning_setting_in_label, icpc2_code_selector, bridging_gap, helm_afr, identical_subject_inputs, measurement_db_embed, proofaug, docs_9a–9g; each with `EVIDENCE.md`, affected-record lists, proposed corrections and an "Ambiguities" section. Patches (3, in the evidence-pack delivery; `audit/patches/` holds the earlier set of the data-quality report: full listing, optional skip of ungraded benchmarks, schema-3 columns): full folder listing without `repo_info`; schema-3

catalogue columns; matching published vectors to current item IDs (builds on the first); each with an offline regression test and `verify_patches.sh` against commit `cc88a959` and current `main`.
Effect on this submission. Findings 3 and 4 define the training exclusions of Section 2; finding 8 is why the predictor groups labels by all eight fields; the other findings concern baseline tooling, documentation or under 1% of the scorable pairs, and the data were used as released. Audited revisions and SHA-256 sums of every delivered file are in `SHA256SUMS` of the evidence-pack delivery sent to the organisers.

F Rerun: commands, hashes, two computation paths

All commands run from the repository root with `PAIEC_WORK` as the only path parameter and `OMP_NUM_THREADS=1`; the Hugging Face token is read from `HF_TOKEN` (the dataset is gated). Step 0 downloads `formatted_tables` at revision `54074453` into a local base and is the only step that needs network access.

```
export OMP_NUM_THREADS=1 PAIEC_WORK=./work   # and HF_TOKEN
make step0 daten    # 0: download_data.py, build_basis.py, daten.py (aggregation)
make tab-cv1 tab1   # 1: Design A phase 1+2 (seeds 0-2, fold seed 1234), CV tables
make fig            # 2: figures of the cross-validation
make smoke          # tests + hash check + fold 0 + comparison to reference, 1e-9
```

The confirmatory runs (seeds 200–202), the method league and this report's figures and tables were produced with the scripts in `konfirmation/`, `liga/` and `report/src/` (see their README files); they are not wired into `make`, and their result files (`konfirmation/hypothesen_tafel.json`, `liga/ergebnisse/`, `report/src/daten/`) are shipped in the repository. Seeds: discovery 0, 1, 2 (fold assignment 1234, pair cap 30, cap seed 0); confirmatory 200, 201, 202 (fold assignment 200); league inner tuning 100; bootstraps 200. Completion is checked mechanically (`status.json` reports 10 finished folds; `auswertung.json` exists; the package check returns 0). Package: `paiec_v2core_final.zip` (32,379 bytes; sha256 `3274b09524221ac268be41f8b52fc56c a341daf72cf9734561be00e0f3a1e331`; sha256 prefix of the embedded canonical state `3c46e5bd4496d360`); a rebuild differs in the embedded build time, so its checksum is not expected to match. Platform check: one formative Codabench evaluation of this package (submission 1008180; at most 1,000 subject–item pairs newly sampled from the hidden test data) returned Brier ALC 0.2056, inside the range of the offline replica (0.1968–0.2129 across regimes, Table 1); a single formative run has no standard error and was not used for any decision. Template: `neurips_2026.sty`, workshop option (`sglblindworkshop,final`). Hardware: 4 CPU cores, 16 GiB RAM, no GPU; aggregation 46 s, training 56 s, one cross-validation design about 3 minutes (4 processes), league about 45 minutes per fold.

Table 9: Two computation paths per core number ("match" = agreement to the stated tolerance). Paths relative to the repository root; the primary audit scripts and the local result folders `cv_REG*`, `cv_V2K*` are not in the repository.

number	path A	path B	match
L10 median 0.0488 (62 benchmarks)	`report/src/analyse/wiederholung.py` on `agg.parquet`, closed-form estimator	`curation/wiederholung_b.py`, pandas on raw responses, `var(ddof=1)`	exact (summaries $\leq 5 \times 10^{-16}$ rel.)
Formula values Eq. 2	`theorie/alc_law.py` (standard library)	`report/src/plots_v2.py` and `tabellen_v2.py` re-implementation from JSON moments	identical (log: `formel_check.log`)
Split-term approximation 0.2130 ALC	`report/src/analyse/split.py` (discovery)	`theorie/alc_law.py --split quadrat`	10^{-6}
EM and BB baselines	`modell/baselines.py` (CV harness)	league harness `liga/liga_methoden.py`	max $\lvert\Delta q\rvert = 0$
v1 path vs. submitted v2.0-core path	`cv_REG1`/`cv_REG2` (v1)	`cv_V2K1`/`cv_V2K2` (v2.0-core)	$\lvert\Delta q\rvert = 0$ on 1,079,568 and 1,214,136 records (phase 2: outside nested Platt stages)
L9 item oracle 0.0065	`report/src/analyse/leiter.py` (test pairs, pair-weighted)	`curation/wiederholung_b.py` (`l9_abgleich`)	0.00649 vs 0.00652
Audit findings 1, 3–9	primary scripts (`check1`–`check6`)	second implementation per finding (different code and data path, same author workflow)	key numbers equal (data-quality report)

Table 10: Data files read by the figure script for each figure (paths relative to the repository root; `daten/` = `report/src/daten/`).

figure	files
1	`modell/ergebnisse/cv_A_p2_S1-S2-S3-S30-kompakt10/auswertung.json (tabelle_alle, vergleiche); floors: daten/beobachtung_discovery.json (U_E), daten/D_exakt.json, daten/leiter.json (L8), curation/wiederholung_b.json (L10); formula: theorie/alc_law.py.`
2	`liga/ergebnisse/vergleich_gepaart.csv (mittel, ki95_unten, ki95_oben, urteil), liga/ergebnisse/meta.json (falten, saaten).`
3	`(a) daten/beobachtung_discovery.json + daten/D_exakt.json via theorie/alc_law.py; konfirmation/hypothesen_tafel.json (H2a, teile.A.S1, teile.B.S1). (b) daten/beobachtung_discovery.json (S1/S3/S30 hier_item).`
4	`daten/leiter.json (leiter_B0), daten/leiter_ergaenzung.json (L2b, L3', L3'', L4', L5'), modell/ergebnisse/cv_A_p2_S1-S2-S3-S30-kompakt10/auswertung.json (BB B0), daten/e3.json (pooling oracle S1 B0), curation/wiederholung_b.json + curation/noise_floor_by_benchmark.csv (L10). Design A, discovery (seeds 0-2).`
5	`daten/verlust.json (alc_anteil_*, gruppe_*, n_benchmarks_schlechter_als_bb), daten/je_benchmark.csv (gewinn_vs_bb, n_subjects, direction).`
6	`curation/fallbeispiele.json (brier_je_budget, floor_R(1-R), diff_hier_item_minus_bb_alc, n_subjects_db).`
7	`daten/verlust.json (alc_anteil_S1/S3, budget_hier_item: sq, itemg, floor); weights 0.1/0.2/0.2/0.2/0.2/0.1.`
8	`modell/ergebnisse/cv_A_p2/zuverlaessigkeit.csv (praediktor, budget, bin, n, q_mittel, R_mittel); regime from modell/ergebnisse/cv_A_p2/auswertung.json primaer_S.`
9	`daten/verlust.json (pooling_sq), daten/beobachtung_discovery.json (S1/beta_binomial).`
10	`curation/noise_floor_by_benchmark.csv (L10_response_weighted, cells_rep, n_median_rep).`

G Reference check

Table 11: Reference check. Of the 32 references cited in this report, the bibliographic record (authors, title, year, venue) was checked against the arXiv record or the DOI record, or the page or repository was accessed, on 20–21 October 2026. The checks were carried out by AI agents under the author's direction (Section "Use of AI assistants") and cover the record, not the full text. A venue given "per arXiv comment" is the one stated in the arXiv record and was not checked at the publisher.

key	check	key	check
Brier [1950]	DOI record checked	Morris [1983]	DOI record checked
Cameron and Miller [2015]	DOI record checked	Murphy [1973]	DOI record checked
Efron and Morris [1975]	DOI record checked	Northcutt et al. [2021]	arXiv record checked
Fox [2010]	DOI record checked	Owen [2024]	arXiv record checked
Gema et al. [2025]	arXiv record + DOI record checked	Pacchiardi et al. [2024]	arXiv record checked
Gneiting and Raftery [2007]	DOI record checked	Platt [2000]	DOI record of the edited volume checked
Guyon et al. [2011]	page or repository accessed	Ruan et al. [2024]	arXiv record checked
Kipnis et al. [2025]	arXiv record checked	Rubin [1976]	DOI record checked
Koren et al. [2009]	DOI record checked	Sawade et al. [2010]	page or repository accessed
Kossen et al. [2021]	arXiv record checked	Stanford AIMS [2026a]	page or repository accessed
Lalor et al. [2016]	DOI record checked	Stanford AIMS [2026b]	page or repository accessed
Lord [2012]	DOI record checked	Stanford AIMS [2026c]	page or repository accessed
MacKay [1992]	DOI record checked	Truong et al. [2025]	arXiv record + proceedings page checked
Madaan et al. [2024]	arXiv record checked	Vivek et al. [2024]	arXiv record + DOI record checked
Maia Polo et al. [2024]	arXiv record + proceedings page checked	Wolpert [1992]	DOI record checked
Miller [2024]	arXiv record checked	Zhou et al. [2025]	arXiv record checked

H Exploratory work, not completed, not part of the submission

Stratum stage. A third level between pair and item (strata of items by visible metadata keys, with the item offset residualised on the stratum) was implemented with 22 unit tests (bit-identical to the submitted model without item features; a deliberate "no shrinkage" failure case triggered). A pilot on one fold ran; the full cross-validation was aborted when the compute budget was reallocated, so no gate decision was possible and no result is reported. **Text-based item prior.** Item texts of 123 benchmarks (114,030 items) were fetched by range requests and embedded with a small sentence encoder on CPU; five hypotheses (within-benchmark R^2 of a ridge on embeddings for item difficulty, cross-benchmark transfer, benchmark location from mean embeddings, ALC effect of a text prior at three subjects between -0.006 and -0.001, runtime limits) were fixed before any evaluation. No evaluation was run before the freeze; nothing is reported, and the submitted predictor uses no text. **Acquisition function.** A coverage-based acquisition policy for the platform's `labeling.py` interface (`akquise/politiken.py`; Appendix A, A.7) was implemented with a simulator of the platform's streaming acquisition (sha256-uniform replica; lemma 1 checked bit-identically on 15,000 candidates; failure cases: broken `labeled` triggers the fallback, a leaked simulator triggers the leak assertion). A pilot on fold 0 fixed the operating characteristic (minimal detectable effect in compact10 0.00034 with 3 seeds); the full run over regimes, orderings and split designs was aborted, so the gate was not evaluated and the platform's random acquisition is used. The pilot numbers in Table 4 are labelled as such.